

CTI Web Services

API Reference Guide

May 2018

The Five9 Computer-Telephony Integration (CTI) Web Services API enable developers to build secure applications for sophisticated call center functionality. The CTI Web Services are part of the Agent Desktop Toolkit, which also contains the softphone Web toolbar for agents.

About Five9

Five9 is the leading provider of cloud contact center software, bringing the power of the cloud to thousands of customers and facilitating more than three billion customer interactions annually. Since 2001, Five9 has led the cloud revolution in contact centers, delivering software to help organizations of every size transition from premise-based software to the cloud. With its extensive expertise, technology, and ecosystem of partners, Five9 delivers secure, reliable, scalable cloud contact center software to help businesses create exceptional customer experiences, increase agent productivity and deliver tangible results. For more information visit www.five9.com.

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What's New

This table lists the changes made in the last six releases of this document:

Release	Changes
May 2018	- Updated the screenshots in Installing the CTI Web Services. - Added this event: loginUIMessage.
August 2017	Corrected an error in the URL of the WSDL.
July 2017	Added a note about PSTN stations to login, loginAsync2, and loginAsync3.
June 2017	- Added this method: isE164Enabled. - Added comment to recordingValue. - In speedDialInfo, replaced code with name. - In userDetails, added sccURL and sccPassword.

Release	Changes
April 2015	- Added these methods: addCrmRecord checkCanCreateConferenceWithAddressType checkCanMakeCallsToAddressType checkCanTransferToAddressType getCrmRecord getCrmRecords lookupCrmRecords removeCrmRecord updateCrmRecord - Added these data types: crmRecord crmRecordLookupResult lookupCursor chatMediaTypeEnabled socialMediaTypeEnabled emailMediaTypeEnabled - Added these faults: CrmRecordNotExistFault DuplicateCrmRecordFault InvalidCrmFieldFault InvalidCrmRecordFault - Added information about upgrading from a previous Five9 release. - Added information about Using Dialing Lists in the E.164 Format. - Added this event: callbackOverdueCleanup

Release	Changes
October 2014	• Added these data types: – agentState – agentStateValue – readyStateValueCommon • Modified setOption. • Added these methods: – checkCanCreateConferenceWith – checkCanMakeCallsTo – checkCanTransferTo – getAgentStateValues – getCurrentWebserver – updateReadyStateCommon • Added these events: – chatBroadcastMessageReceived – modelChanged – readyStateChangedCommon – restartStationStarted – stationDisconnected • Added ServiceFault. • Added these agent permissions: – canManageSkillAvailability – canPickSalesforceObjectForCallLog – canTransferToAgents – canTransferToSkills – canCreateConferenceWithAgents – canCreateConferenceUsingSkills – canConfigureAutoAnswer – canSelectSysRecycleDisposition – isTextEnabled

Chapter 1

About the CTI Web Services API

This guide describes how to use the CTI Web Services API to build secure applications for sophisticated call center functionality.

[Audience](#)

[Agent Desktop Toolkit](#)

[Agent State Model](#)

Audience

This guide is intended for developers who want to create contact center applications. These developers must understand these technologies:

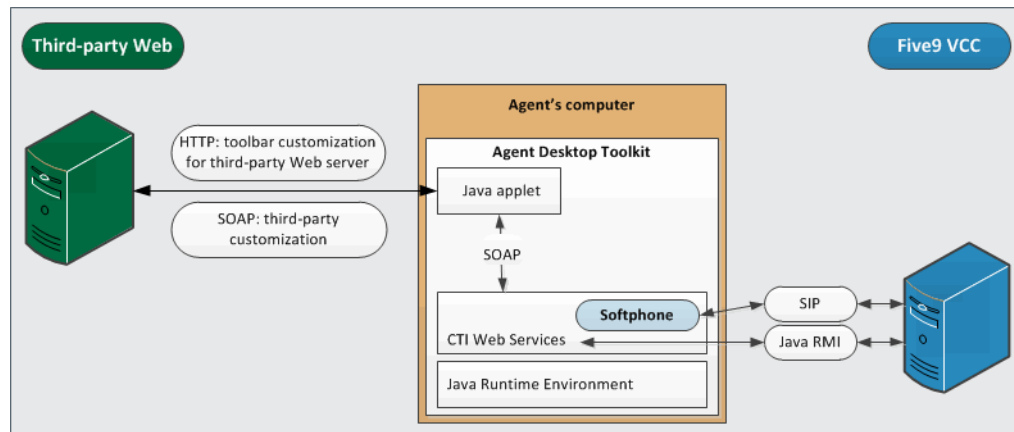
- SOAP
- HTTP
- XML
- JSP, ASP, CGI, or Perl
- Computer-telephony integration concepts, processes, terminology, and applications
- Telephony events, call routing, and agent statuses
- Overall call center integration

Agent Desktop Toolkit

The Five9 Agent Desktop Toolkit comprises the Computer-Telephony Integration (CTI) Web Services and the softphone Web toolbar for agents.



This figure shows how the Agent Desktop Toolkit components that reside on the agent's computer interact with the Five9 VCC and the agent's organization.



The CTI Web Services communicate with the Virtual Call Center (VCC) with the Java Remote Method Invocation (RMI) API, which can send many types of requests to the VCC, for example:

- Logging in and out
- Modifying the agent statuses
- Initiating, answering, and processing all parts of calls
- Processing voicemail messages and callbacks
- Assigning call dispositions
- Distributing events to the CTI Web Services clients

CTI Web Services

The CTI Web Services API enable your client to communicate with the VCC by using an XML-encoded SOAP interface. After you have installed the Agent Desktop Toolkit on your computer, you can obtain the XML documents at these URLs:

- WSDL: <http://localhost:8080/agent/v2?wsdl>
- XSDs:
 - Methods: http://localhost:8080/agent/agent_v2.core.xsd
 - Events: http://localhost:8080/agent/agent_v2.events.xsd
 - Data types: http://localhost:8080/agent/agent_v2.types.xsd

The CTI Web Services support the cross-origin resource sharing (CORS) standard with JavaScript, Silverlight, and Flash.

Five9 Softphone Web Toolbar

The softphone Web toolbar for agents is a browser plug-in that enables you to test your application by processing calls with a softphone, PSTN, or gateway. In addition to Five9 call center functionality, the toolbar provides advanced telephony controls:

- Login access
- Agent statuses
- Call dispositions
- Call logs
- Call recording
- Access to voicemail and callbacks
- Scripts, worksheets, and connectors

You can customize the toolbar to work with other CTI Web Services clients, and you can register the toolbar to receive Five9 VCC events so that the toolbar status is updated after events by other CTI Web Services clients.

The toolbar is optional. If you prefer, you can create your own application to handle telephony controls and call center functionality.

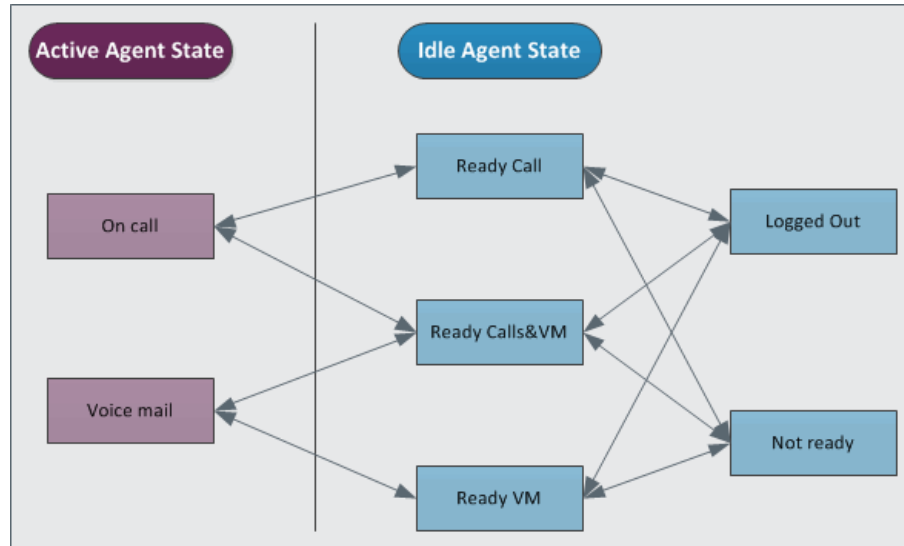
For complete information about the Web toolbar for agents, see the [Agent Desktop Toolkit Guide](#).

Agent State Model

When logged into a station, agents can select among these states to process calls, voicemail messages, or both:

- Ready Call
- Ready VM
- Ready Call&VM

The VCC administrator determines which states agents can select. This figure shows the options open to agents when not processing a call or voicemail message.



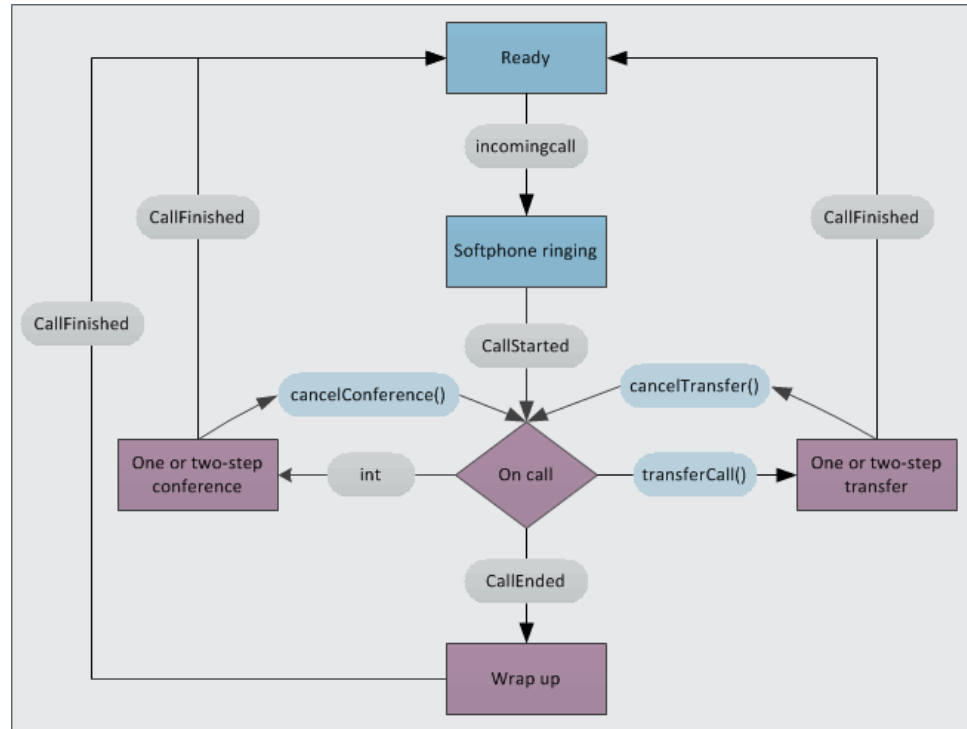
An agent is active when taking a call or listening to voicemail messages. An agent can have only one active call at a time. The agent can transfer, place on hold, park, and create a conference call by adding one or more participants. The available options depend on those allowed by the administrator.

After finishing a task, agents assign one of the dispositions enabled by the VCC administrator:

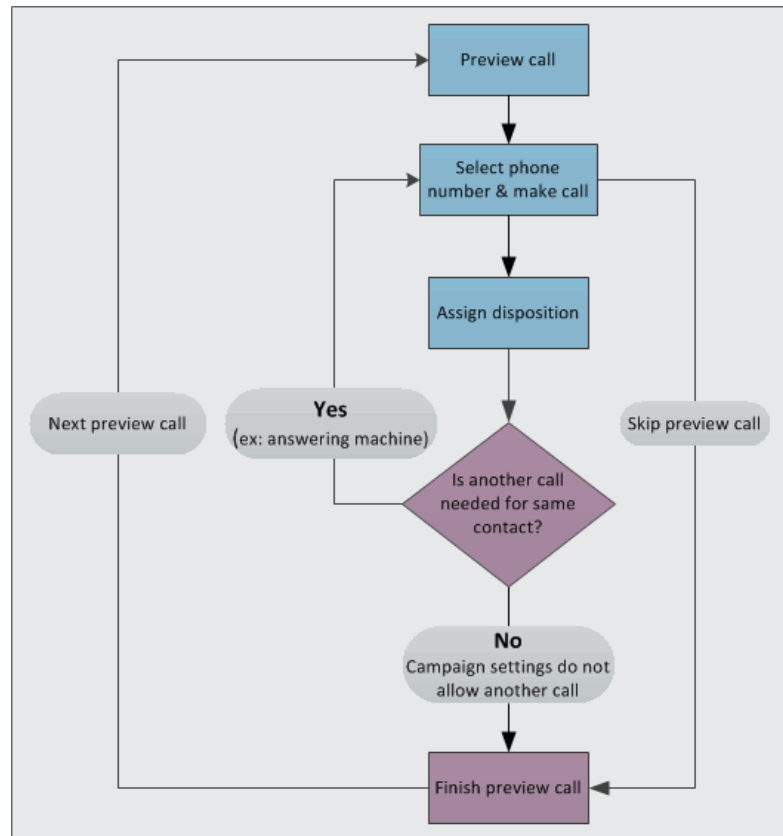
- System dispositions: default VCC dispositions. System dispositions can be used, for example, for parked and transferred calls.
- Custom dispositions: the administrator can define as many dispositions as needed and associate them with a campaign.

A call is sent to an agent or to the Web Services client as follows:

- Incoming call: incoming call campaign to the VCC.



- Outbound call:
 - Call or callback initiated by an agent.
 - Call initiated by the Five9 dialer. The call is sent to an agent when the connection is established or a contact is reached.
 - Call initiated in preview mode by the Five9 dialer. The call is sent to an agent after the agent has accepted the Preview Call. When a call preview is presented to an agent, one or more phone numbers, such as business, mobile, and home, may appear for the contact. The agent can call any number or can skip the contact. The figure below shows the Preview call flow.



Chapter 2

Installing and Using the Five9 CTI Web Services

This chapter describes how to install and use the CTI Web Services:

[Installing the CTI Web Services](#)

[Using the CTI Web Services](#)

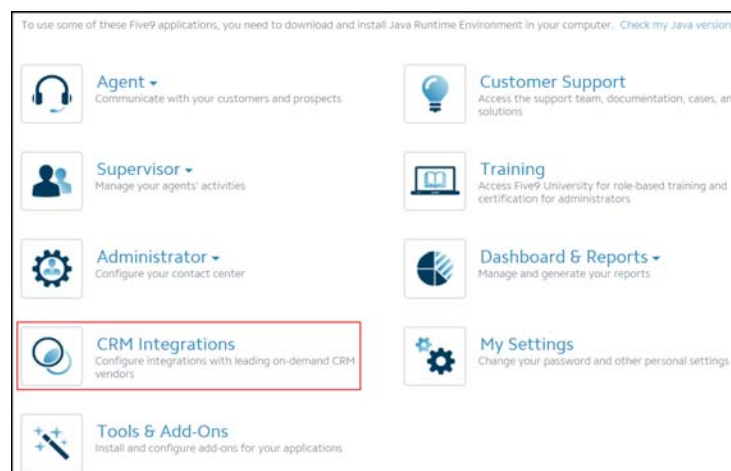
[Removing the Adapter](#)

Installing the CTI Web Services

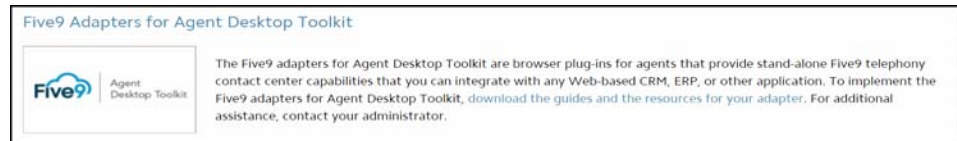
For information about the supported browsers and Java Runtime Environment, see the [VCC Technical Requirements](#).

Important If you are currently using a previous version of the Five9 Agent Desktop Toolkit adapter, you need to stop the adapter, but you do not need to uninstall it. Download and install the new adapter when your administrator instructs you to do so.

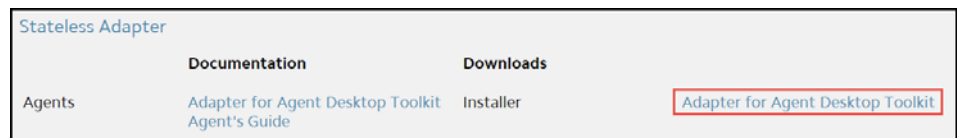
- 1 Log into your Five9 account.
- 2 Click **CRM Integrations**.



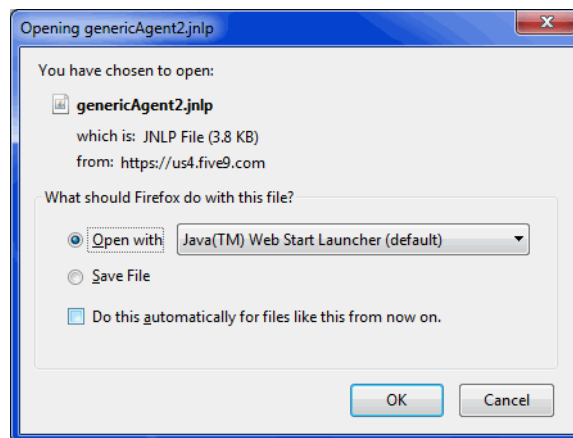
3 Click **Five9 Adapters For Agent Desktop Toolkit**.



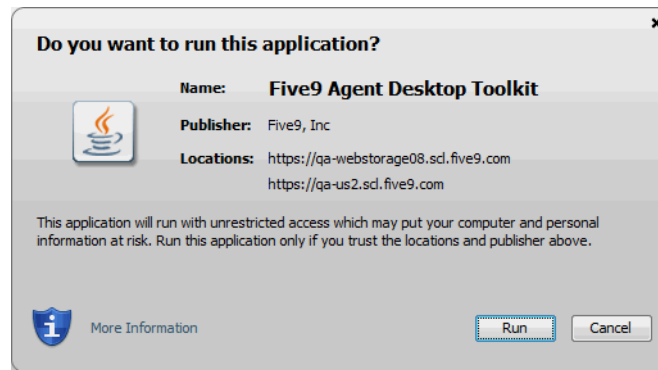
4 Scroll down to the **Stateless Adapter** section and click **Adapter for Agent Desktop Toolkit**.



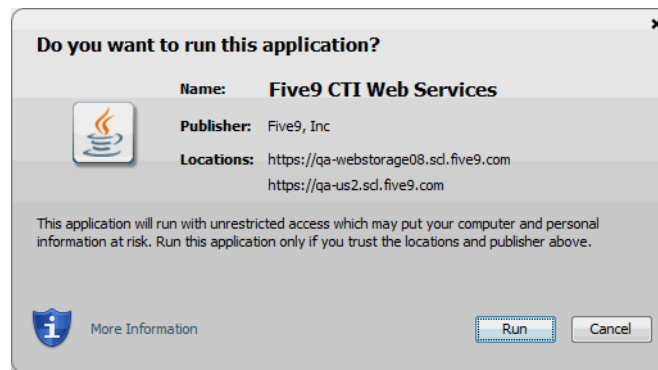
5 Open or save the file, and click **OK**.



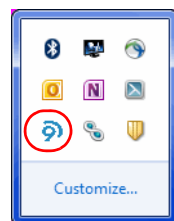
- 6 To install the Five9 Agent Desktop Toolkit, click **Run**.



- 7 To install the Five9 CTI Web Services, click **Run**.



When the CTI Web Services have started, the Five9 logo appears in the system tray.

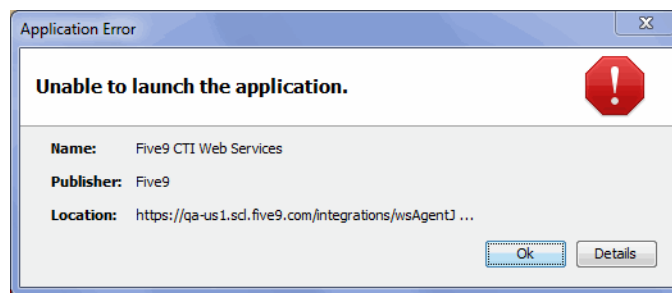


Using the CTI Web Services

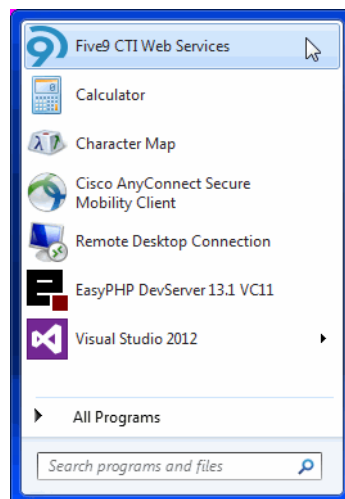
By default, the CTI Web Services starts automatically when you start your computer. In this case, you see the Five9 logo in the system tray. You must have an Internet connection to start and use the CTI Web Services.

Starting the CTI Web Services

If the CTI Web Services cannot start automatically, you see this error message.

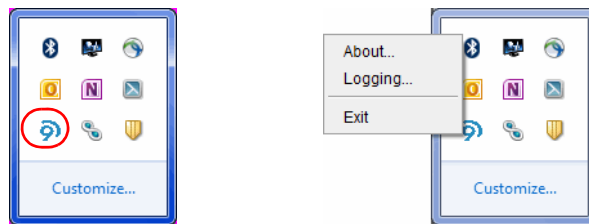


To start the CTI Web Services manually, click **Start > All Programs > Five9 > Five9 CTI Web Services**.



Stopping the CTI Web Services

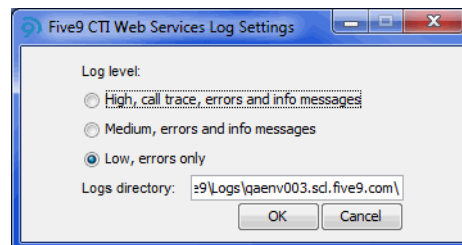
To stop the CTI Web Services, right-click the Five9 icon in the system tray and choose **Exit**.



Defining Log Settings

The logs may be requested by Customer Support to troubleshoot issues with the Five9 CTI Web Services.

- 1 Right-click the icon in the system tray, and select **Logging**.
- 2 Select your options:



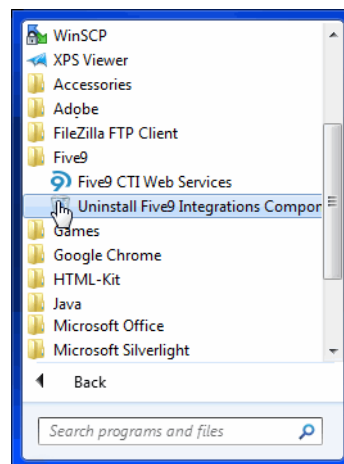
- Log level: types of messages to log. In production, set the log level to **Low**. When debugging, set the log level to **High**.
 - Logs directory: location for the generated log files. The Five9 Web Services log file is named adapter . log. The Five9 toolbar log file is named applet . log.
- 3 Click **OK**.

Removing the Adapter

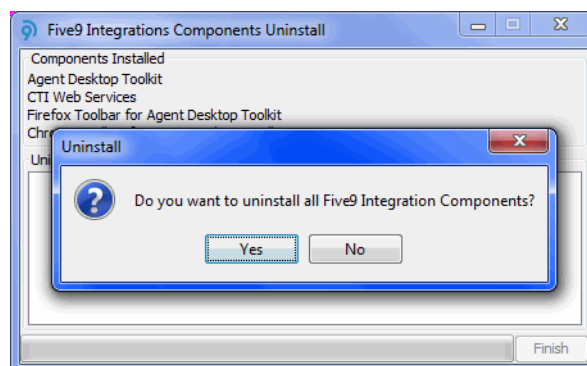
If you need to completely remove the components of the Agent Desktop Toolkit, follow these steps.

Warning Use this procedure only if directed to do so by your administrator. Do not use this procedure to update your adapter.

- 1 Stop the adapter.
- 2 Remove the integration:
 - a Click **Start > All Programs > Five9 > Uninstall Five9 Integrations Components**.



The integration components window appears.

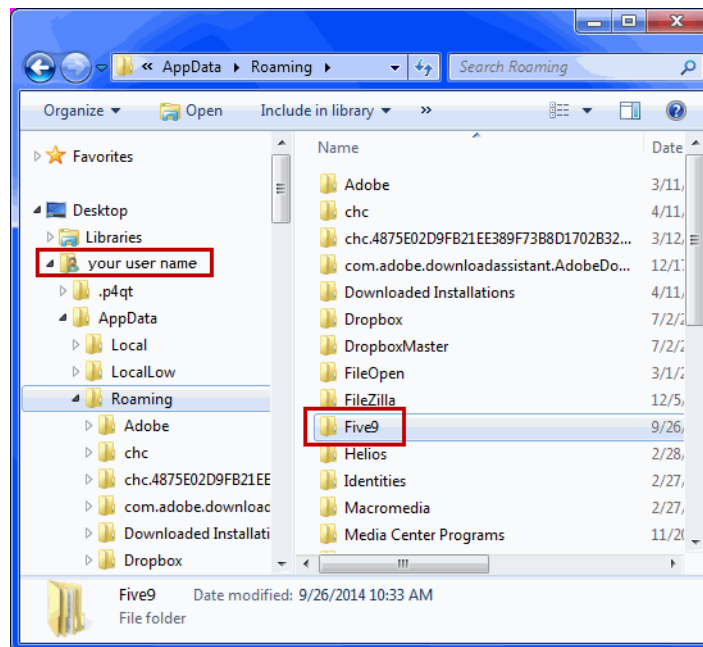


- b Click **Yes**.
 - c When done, click **Finish**.

3 Navigate to `<your user name>\AppData\Roaming`.

4 Delete the Five9 folder completely.

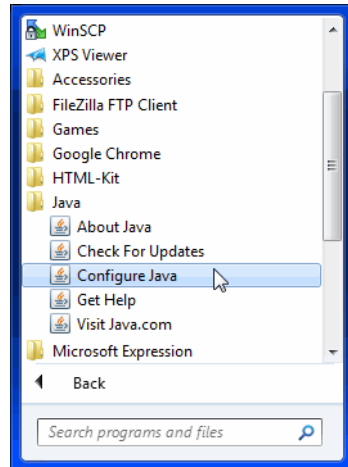
Note You may need to reboot before being able to delete this folder.



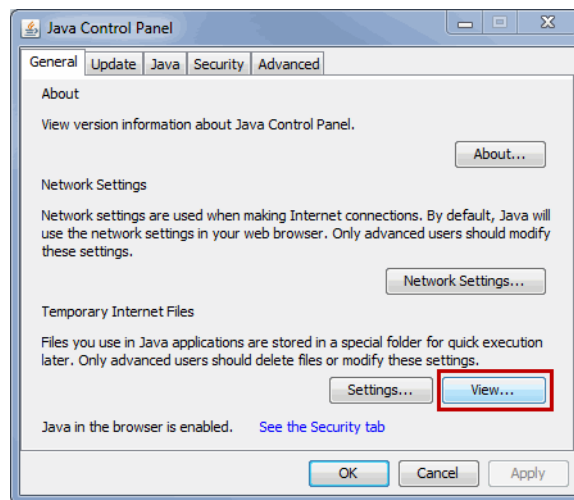
5 Locate the Java Control Panel.

The location of the panel depends on your platform and on the Java version installed on your computer. The steps below apply to Java 7 Update 40 (7u40) and later versions on Windows. If you use a lower Java version or the Mac platform, click one of these links: [Windows](#) or [Mac](#) for instructions.

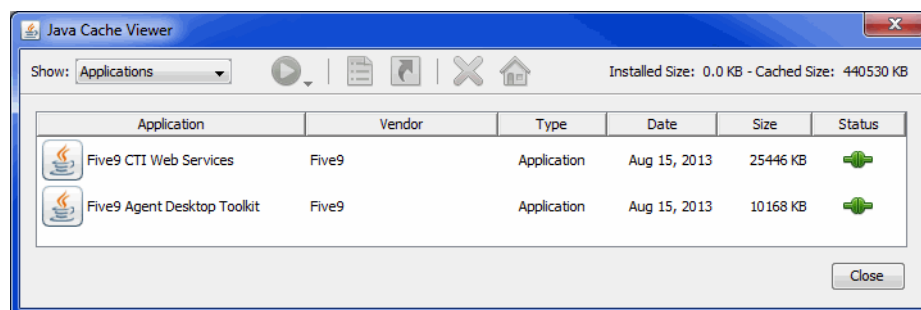
6 Click **Start > All Programs > Java > Configure Java**.



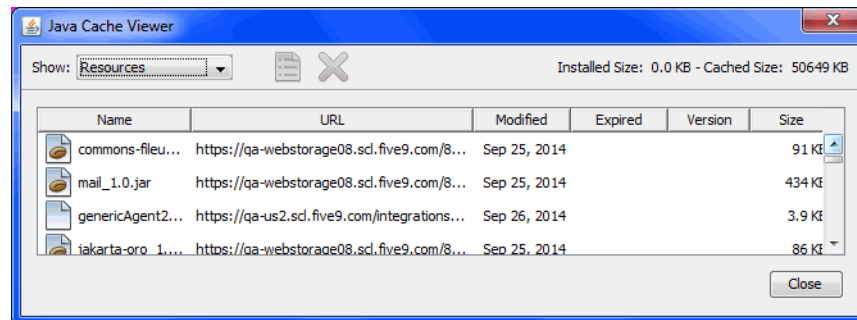
- 7 On the General tab of the Java Control Panel, click **View** at the bottom.



- 8 In the Java Cache Viewer, highlight the two components, and click the red **X** or **Delete** on your keyboard.



- 9 In the Show menu, select **Resources**.



- 10 Select (CTRL-A) and delete all resources.

Chapter 3

Methods

The methods are located at http://localhost:8080/agent/agent_v2.core.xsd.

[Agent States](#)

[Audio Files](#)

[Call Variables](#)

[Call Features](#)

[Callbacks](#)

[Chat Sessions](#)

[Connectors](#)

[Contact Records](#)

[CTI Web Services](#)

[Dispositions](#)

[DNC List](#)

[Events](#)

[Greetings](#)

[Scripts](#)

[Skill Prompts](#)

[Softphone Options](#)

[User Account](#)

[User Information](#)

[VCC Configuration](#)

[Voicemail](#)

[Worksheets](#)

Agent States

This section describes how to set and modify the state of an agent.

[getAgentStates](#)

[getAgentStateValues](#)

[getNotReadyReasonCodes](#)

[setNotReadyState](#)

[setReadyState](#)

getAgentStates

Use this method to obtain a list of agent states.

getAgentStates. This method contains no parameters.

getAgentStatesResponse.

Parameter	Type	Description
return	agentState [0 . . unbounded]	List of agent states.

getAgentStateValues

Use this method to obtain a list of configured agents and their current agent state. You may use this information, for example, to determine the state of agents in the call center before assigning calls, transfers, and conferences to them.

getAgentStates. This method contains no parameters.

getAgentStatesResponse.

Parameter	Type	Description
return	agentStateValue [0..unbounded]	List of possible agents' states.

getNotReadyReasonCodes

Use this method to obtain the list of *Not Ready* reason codes.

getNotReadyReasonCodes. This method contains no parameters.

getNotReadyReasonCodesResponse.

Parameter	Type	Description
return	reasonCodeValue [0..unbounded]	List of available <i>Not Ready</i> reason codes and information about each code.

setNotReadyState

After you have obtained the list of reason codes, use this method to change the agent state to Not Ready.

setNotReadyState.

Parameter	Type	Description
reasonId	long	Reason ID.

emptyResponse.

setReadyState

Use this method to change the ready state of an agent. If this method is used during a call, the state is changed only after the call is finished.

setReadyState.

Parameter	Type	Description
stateId	long	ID received from the readyStatesUpdated event.

emptyResponse.

Audio Files

This section contains methods that you can use manage sound files.

[getAudioPlayerVolume](#)
[pauseAudioPlaying](#)
[playFile](#)
[resumeAudioPlaying](#)
[setAudioPlayingPosition](#)
[setAudioPlayerVolume](#)
[stopAudioPlaying](#)

getAudioPlayerVolume

Use this method to obtain the current volume of the internal audio player, which is used to play reminders.

getAudioPlayerVolume. This method contains no parameters.

getAudioPlayerVolumeResponse.

Parameter	Type	Description
return	float	Volume value: 0.00 – 1.0.

pauseAudioPlaying

Use this method to pause the audio player, for example during voicemail playing.

pauseAudioPlaying. This method contains no parameters.

emptyResponse.

playFile

Use this method to play audio files when implementing notification features.

playFile.

Parameter	Type	Description
filename	string	Path to the audio file.

emptyResponse.

resumeAudioPlaying

After using [pauseAudioPlaying](#), use this method to resume playing the file, such as voicemail.

resumeAudioPlaying. This method contains no parameters.

emptyResponse.

setAudioPlayingPosition

Use this method to set the position of the internal audio player.

setAudioPlayingPosition.

Parameter	Type	Description
position	int	Position in percentage: 1 – 100.

emptyResponse.

setAudioPlayerVolume

Use this method to set the volume of the internal audio player, which is used to play reminders.

setAudioPlayerVolume.

Parameter	Type	Description
volume	float	Volume value: 0.00 – 1.0.

emptyResponse.

stopAudioPlaying

Use this method to stop the audio player.

stopAudioPlaying. This method contains no parameters.

emptyResponse.

Call Variables

This section contains methods that you can use to manage call variables.

[getCallAttachedVariables](#)

[getCAVFieldViews](#)

[getForceCavViews](#)

[setCallAttachedVariable](#)

getCallAttachedVariables

Use this method to obtain an array of call variables.

getCallAttachedVariables. This method contains no parameters.

getCallAttachedVariablesResponse.

Parameter	Type	Description
return	cavFullValue [0..unbounded]	Array of call variables.

getCAVFieldViews

Use this method to obtain a list of call variables fields.

getCAVFieldViews. This method contains no parameters.

getCAVFieldViewsResponse.

Parameter	Type	Description
return	cavViewFullValue [0..unbounded]	List of call variables fields.

getForceCavViews

Use this method to find out whether the agent is forced to view the call variables.

getForceCavView. This method contains no parameters.

getForceCavViewResponse.

Parameter	Type	Description
return	boolean	Whether the agent is forced to view the call variables.

setCallAttachedVariable

Use this method to change the values of call variables at any time during a call.

setCallAttachedVariable.

Parameter	Type	Description
id	long	Variable ID provided by getCallAttachedVariables.
value	string	New value for call variable.

emptyResponse.

Call Features

This section contains the methods that you can use to start, process, and end calls.

[Click-to-Call](#)
[Conferences](#)
[Disconnecting](#)
[Holding](#)
[Parking](#)

[Previewing](#)
[Recording](#)
[Speed Dial](#)
[Transfers](#)

Click-to-Call

[getAvailableCampaigns](#)
[makeCall](#)
[makeTestCall](#)
[startClick2Dial](#)

Using Dialing Lists in the E.164 Format

If your domain is not enabled for the E.164 format for international phone numbers, your agents must enter 10 digits to conform to the [North American Numbering Plan](#). This section does not apply to you.

If your domain is enabled for the E.164 format, your agents need to enter phone numbers for all calls according to your configuration: in your national format or up to 15 digits, starting with the plus (+) sign and the country code.

The E.164 format is used to route calls, process lists, and display numbers in your applications and reports. When you upload a list, all non-numeric characters, such as dots, dashes, spaces, and parentheses, and the country prefix are removed. Numbers are stored in the E.164 format, starting with the plus sign.

You may also configure in the VCC a national phone number format. In this case, you can upload lists that contain numbers in E.164 format, in your national format, or both. If you do not specify a format, the VCC uses the E.164 format for all numbers. For more information, see the [Basic Administrator's Guide](#). For example, if you operate in the UK, these numbers are processed as follows:

Phone number format	How the number is processed
International format	1 Belgium phone number is added to a list: 0032.20.3456.7899 2 Phone number is processed in VCC: International prefix (00) and non-numeric characters are removed. Plus sign is added. 3 Stored number: +322034567899
National format	1 UK phone number is added to a list: 02034567899 2 Phone number is processed in VCC: National prefix (0) and non-numeric characters are removed. UK country code and the plus sign are added. 3 Stored number: +442034567899
No specified format	Number stored in E.164 format.

getAvailableCampaigns

Use this method to find the ID of the campaign that you need to add to `makeCall`.

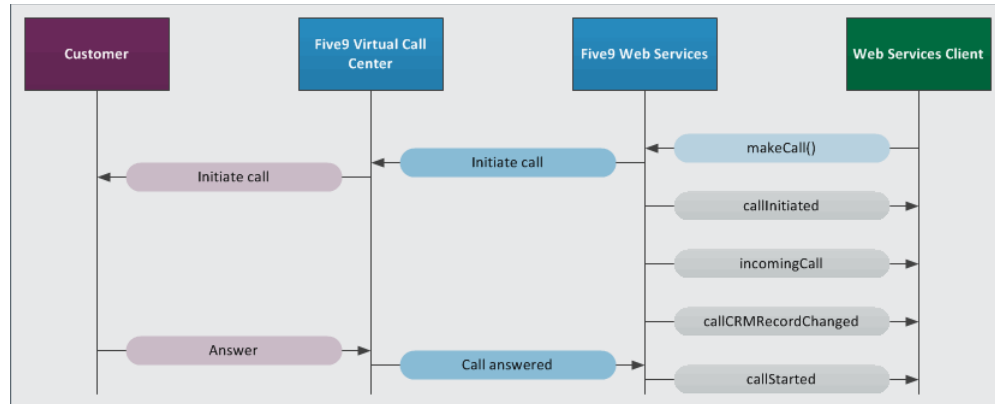
getAvailableCampaigns. This method contains no parameters.

getAvailableCampaignsResponse.

Parameter	Type	Description
return	campaignInfo	List of available campaigns.

makeCall

Use this synchronous non-blocking method to initiate a manual call. See also the [startClick2Dial](#) method. This figure shows the events that occur with this method. If you are enabled to use the E.164 format for international telephone numbers, see [Using Dialing Lists in the E.164 Format](#) before using this method.



When an error occurs at the beginning or during a call, the [makeCallFailed](#) event is sent to the client. When the call is completed, the [callEnded](#) event is sent to the client.

makeCall.

Parameter	Type	Description
number	string	10 unformatted digits that represent a phone number, agent name, skill name, extension, or campaign ID. All non-numeric characters, except the + sign at the beginning of the phone number (+19998887777) are removed.
campaignId	long	Campaign assigned to the call.
checkDnc	boolean	Whether to check the called number against the Do-Not-Call list.
callbackId	long	ID of the callback associated with the call.

emptyResponse.

makeTestCall

This method is not currently used.

startClick2Dial

Use this method to start a [click2DialInitiated](#) event to pass a call number to the client but not start a call yet. For example, the client would prompt the agent to select a

campaign. If the agent knows the campaign ID, use `makeCall` instead. If you are enabled to use the E.164 format for international telephone numbers, see [Using Dialing Lists in the E.164 Format](#) before using this method.

`startClick2Dial.`

Parameter	Type	Description
<code>arg0</code>	string	Number delivered by the <code>click2DialInitiated</code> event. All non-numeric characters, except the + sign at the beginning of the phone number (+19998887777) are removed.

`emptyResponse.`

Conferences

Use these methods to create and manage conference calls.

addToConference	leaveConference
cancelConference	leaveConference2
checkCanCreateConferenceWith	leaveConference3
completeConference	parkConferenceParticipant
conferenceIn	removeConferenceParticipant
disconnectCall	

When errors occur during a conference, the [conferenceError](#) event contains a message that describes the cause.

`addToConference`

Use this method to add a participant to an existing conference. Conference participants can be added in two ways:

- Warm: the agent speaks to the third party before completing the conference.
- Mute: the third party is added immediately.

If a participant is added successfully, the [conferenceParticipantAdded](#) event contains information about the participant.

To start a conference, see [conferenceIn](#).

addToConference.

Parameter	Type	Description
callSessionId	string	Unique call identifier.

emptyResponse.**cancelConference**

Use this method to cancel the second step of a warm conference.

cancelConference. This method contains no parameters.

emptyResponse.**checkCanCreateConferenceWith**

Use this method to check if the agent can create conference of specified type.

checkCanCreateConferenceWith.

Parameter	Type	Description
type	int	Type of call: • 0: Can create conference with external number. • 1: can create conference with agent. • 2: can create conference with skill.

checkCanCreateConferenceWithResponse.

Parameter	Type	Description
return	boolean	Whether the agent has the required permission.

completeConference

Use this method to perform the second step of a warm conference. If the conference is successfully created, the [conferenceCompleted](#) event is sent.

completeConference. This method contains no parameters.

emptyResponse.

conferenceIn

Use this method to start a conference. Agents can start a conference if they have permission to do so. If an error occurs while creating a conference, the [conferenceCreateFailed](#) event describes the failure.

conferenceIn.

Parameter	Type	Description
numberToConference	string	External number, user name, or campaign.
isWarmTransfer	boolean	Whether participant should be added with a 2-step process.
checkDnc	boolean	Whether to check the DNC list.

emptyResponse.

disconnectCall

Use this method to terminate a conference, which disconnects all participants. This is the same method that you use to terminate a call except that, in this case, the method returns nothing.

disconnectCall. This method contains no parameters.

emptyResponse.

leaveConference

Use this method to leave the conference without affecting the remaining parties in the conference.

leaveConference.

Parameter	Type	Description
dispositionId	long	Disposition ID.

emptyResponse.

leaveConference2

Use this method to enable a user to leave the conference without affecting the remaining parties in the conference.

leaveConference2.

Parameter	Type	Description
dispositionId	long	Disposition ID.
redialTimeout	long	Minimum time before redialing the conference number.

emptyResponse.

leaveConference3

Use this method to enable a user to leave the conference without affecting the remaining parties in the conference.

leaveConference3.

Parameter	Type	Description
dispositionCallId	string	Disposition ID already assigned to the call.

emptyResponse.

parkConferenceParticipant

Use this method to park a conference participant.

parkConferenceParticipant.

Parameter	Type	Description
callSessionId	string	ID of the parked call.

emptyResponse.

removeConferenceParticipant

Use this method to remove a participant from the conference. All participants, except the first participant in the call, can be removed. After a participant is removed from the conference, the [conferenceParticipantRemoved](#) event is sent.

removeConferenceParticipant.

Parameter	Type	Description
callSessionId	string	Session ID of the participant.

emptyResponse.

Disconnecting

This section describes how to terminate or reject a call.

disconnectCall

Ending a call is a two-step process:

- 1 Use this method to end a call. The agent's now has some time to complete any after-call work.
- 2 Although the method terminates the connection, the call is not completed until the agent assigns a disposition. Follow up with the [finishCall](#) or [finishCall2](#) method.

disconnectCall. This method contains no parameters.

emptyResponse.

rejectCall

Use this method to reject the current call. If the agent disables the auto answer feature, the client sends the [incomingCallInfo](#) event. At that time, the client can reject the call.

rejectCall. This method contains no parameters.

emptyResponse.

Holding

A call can be placed on hold at any time.

toggleHold

Use this method to place a call on hold or to retrieve it. To confirm that the operation is successful, the `callOnHold` and `callOffHold` events are sent.

toggleHold.

Parameter	Type	Description
flag	boolean	True to place on hold, false to take off hold.

emptyResponse.

Parking

The methods in this section describe how to manage parked calls.

[disconnectParkedCalls](#)

[parkCall](#)

[retrieveCall](#)

disconnectParkedCalls

This method is not currently used.

parkCall

Use this method to park a call that can later be retrieved. After a call is parked, the [callParked](#) event is sent. In case of error, the [parkError](#) event is sent. When a call is parked, the agent can initiate or receive another call.

parkCall. This method contains no parameters.

emptyResponse.

retrieveCall

Use this method to retrieve a parked call. After retrieving the call, the [incomingCall](#) event is sent.

retrieveCall.

Parameter	Type	Description
callSessionId	string	ID of the parked call.

emptyResponse.

Previewing

In the VCC, you can set the preview dialing mode for an outbound campaign. In that mode, the dialer shows a preview of the call to the agent before dialing the call. The preview contains information about the contact, such as the home, work, and mobile phone numbers. The agent or the client can choose the phone number to dial or can choose to skip the contact.

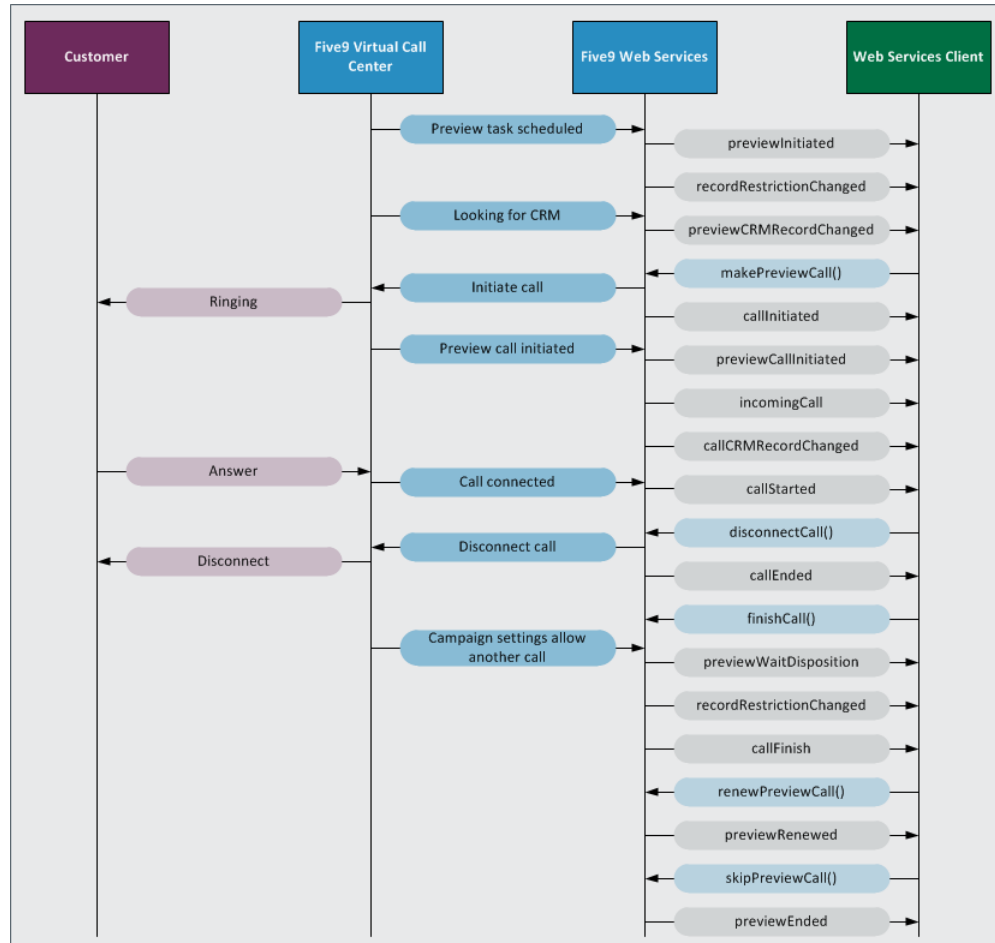
[continuePreview](#)
[endPreviewCall](#)
[getPreviewSkipDispositions](#)
[getPreviewSkipDispositions2](#)
[makePreviewCall](#)
[skipPreviewCall](#)
[skipPreviewCall2](#)

The call initiation process is divided into two parts:

Call presentation: The Five9 VCC sends to the agent the contact information and the phone numbers that can be dialed. The agent selects the phone number to dial or skips the contact. The call to the number selected by the agent may not be successful, such as when an answering machine is detected. Depending on the configuration of the outbound campaign, this same contact may be presented again to the agent, but without the previously called number. Instead, the agent must select among the remaining numbers.

The preview call task cycle is started with the `previewInitiated` event, which contains the [callInfo](#) structure. After this event is sent, the client knows that a preview call is scheduled for that agent by the VCC. The next event is [previewCRMRecordChanged](#), which contains the [map](#) structure with contact database records for the contact presented in the preview call. After the Preview Call task is finished the [previewEnded](#) event is sent.

This figure shows the flow of preview call events. A preview call is sent to an agent who dials one of the phone numbers shown in the preview call. If the agent reaches an answering machine, the call preview is renewed. The contact is shown again, with the list of the remaining phone numbers. In this case, the agent chooses to skip that contact.



Use the following methods after the client receives the [previewInitiated](#) event.

continuePreview

After the client receives the [previewWaitDisposition](#) event, use this method to decide whether to continue previewing a contact record when a call is unsuccessful, such as when an answering machine is detected or no one answers. This option is possible if campaign settings allow agents to make another call to the same record. The client can either allow the preview to continue (`dialAgain=True`) or assign a disposition (`dialAgain=False`).

continuePreview.

Parameter	Type	Description
<code>dispositionId</code>	long	Disposition ID.

dialAgain	boolean	Whether to continue previewing a record or to proceed to the next record.
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emptyResponse.

endPreviewCall

Use this method to end previewing a call.

endPreviewCall.

Parameter	Type	Description
dispositionId	long	Disposition ID obtained from getCampaignDispositions .

emptyResponse.

getPreviewSkipDispositions

Use this method to obtain a list of call dispositions from which the agent can choose when skipping a call preview. To enforce a time-out when selecting a disposition, see [getPreviewSkipDispositions2](#).

Important In a preview outbound campaign, you can use this method only after you receive the `previewInitiated` event because the CTI Web Services retrieve the correct dispositions from the VCC for the campaign to which the call belongs. If you use this method before the agent sees the preview, you receive an error because the dispositions have not yet been retrieved.

Be sure that agents log into the VCC with skills used by the preview campaign. Otherwise, the VCC does not deliver preview calls to the agents.

getPreviewSkipDispositions. This method contains no parameters.

getPreviewSkipDispositionsResponse.

Parameter	Type	Description
return	dispositionValue [0..unbounded]	List of available dispositions for skipPreviewCall .

getPreviewSkipDispositions2

Use this method to obtain a list of call dispositions from which the agent can choose when skipping a call preview.

getPreviewSkipDispositions2. This method contains no parameters.

getPreviewSkipDispositions2Response.

Parameter	Type	Description
return	dispositionValue2 [0..unbounded]	List of available dispositions, including system dispositions, and a time-out parameter when selecting a disposition code.

makePreviewCall

Use this method to dial a call that can be previewed. When the method is successful, the [previewCallInitiated](#) event, which contains the session ID of the call, is sent.

makePreviewCall.

Parameter	Type	Description
number	string	Phone number selected by the agent.

emptyResponse.

renewPreviewCall

This method is no longer used. See [continuePreview](#) instead.

skipPreviewCall

Use this method to skip a call preview.

skipPreviewCall.

Parameter	Type	Description
dispositionId	long	One of the campaign dispositions obtained from getPreviewSkipDispositions .

emptyResponse.

skipPreviewCall2

Use this method to skip a call preview.

skipPreviewCall2.

Parameter	Type	Description
dispositionId	long	One of the campaign dispositions obtained from getPreviewSkipDispositions .
timeout	long	Maximum time for assigning the disposition.

emptyResponse.

Recording

Use this method to manage call recordings.

recordCall

Use this method to start and finish a call recording.

recordCall.

Parameter	Type	Description
flag	boolean	True to start; false to stop recording.

emptyResponse.

Speed Dial

Use this method to obtain speed dial information.

getSpeedDials

Use this method to obtain the list of internal resources (agents, supervisors, administrators, and skills) than an agent can reach with speed dial.

getSpeedDials. This method contains no parameters.

getSpeedDialsResponse.

Parameter	Type	Description
return	speedDialInfo [0..unbounded]	List of information about speed dial numbers.

Transfers

[cancelTransfer](#)

[checkCanTransferTo](#)

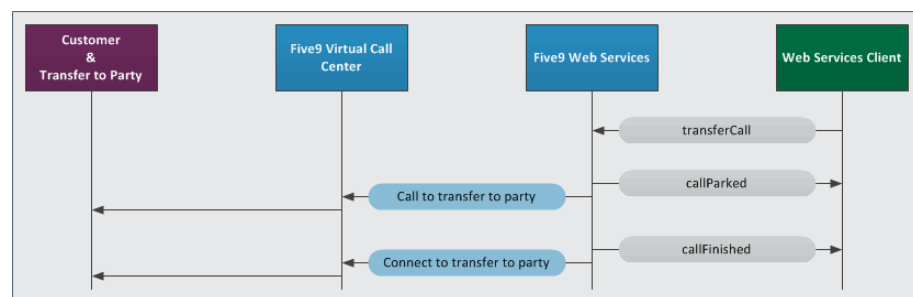
[completeTransfer](#)

[transferCall](#)

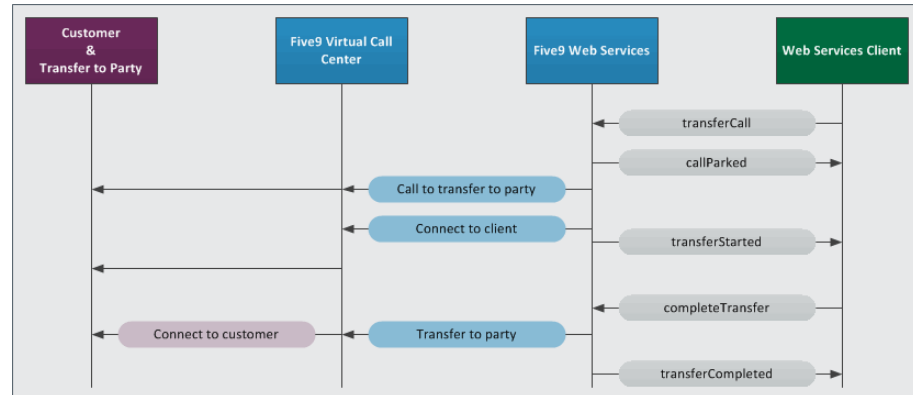
[transferCall2](#)

Transfers can be mute or warm:

- During a mute transfer, when the second call leg is not successful, the party being transferred is parked. If an error occurs during transfer, such as when the phone number is invalid, the `TransferError` event is sent to the client. This figure shows the flow of a mute call transfer.



- During a warm transfer, the call can be disconnected by the remote party. In this case, the `transferCallDisconnected` event is sent to the client. When the consultant disconnects the call, the `consultCallDisconnected` event is sent to the client. After the consultant is disconnected, the transferred party is put on hold. A warm transfer applies only to the current active voice. This figure shows the flow of a warm call transfer.



cancelTransfer

Use this method to cancel a warm (two-phase) transfer before using the `completeTransfer` method, for example when the call is parked.

cancelTransfer. This method contains no parameters.

emptyResponse.

checkCanTransferTo

Use this method to check if the agent can transfer to specified type.

checkCanTransferTo.

Parameter	Type	Description
type	int	Type of call: 0: Can transfer to external number. 1: can transfer to agent. 2: can transfer to skill.

checkCanTransferToResponse.

Parameter	Type	Description
return	boolean	Whether the agent has the permission.

completeTransfer

Use this method to perform the second step of a warm transfer. If the transfer is successful, the [transferCompleted](#) event is sent.

completeTransfer. This method contains no parameters.

emptyResponse.

transferCall

Use this method to transfer the current active or parked call. This method enables you to check the DNC list.

transferCall.

Parameter	Type	Description
callSessionId	string	ID of the call to transfer.
number	string	External number, agent name or extension, or skill name.
timeout	int	Timeout before disconnection.
dispositionId	long	ID of the disposition selected from the list received from getCampaignDispositions .
warm	boolean	Whether the transfer is warm (with consultation) or mute (single-step transfer). If set to true, complete the transfer with completeTransfer .
checkDnc	boolean	Whether to check the number on the DNC list.

emptyResponse.

transferCall2

Use this method to transfer the current active or parked call. This method enables you to set the minimum time before redialing the number.

transferCall2.

Parameter	Type	Description
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callSessionId	string	ID of the call to transfer.
number	string	External number, agent name or extension, or skill name.
timeout	int	Timeout before disconnection.
dispositionId	long	ID of the disposition selected from the list received from getCampaignDispositions .
warm	boolean	Whether the transfer is warm (with consultation) or mute (single-step transfer). If set to true, complete the transfer with completeTransfer .
redialTimeout	long	Minimum time before redialing the transfer number.

[emptyResponse](#).

Callbacks

Callbacks can be scheduled for a specific agent or for a skill.

addCallback	getCRMForCallbackId
addCallback2	getCRMMap
addCallback3	rejectLiveCallback
answerLiveCallback	removeCallback
checkAndFilterCallback	resetRemindedCallback
dismissCallback	snoozeCallback
getCallbacks	updateCallback

When callbacks are changed by the client or the supervisor, the [callbacksChanged](#) event is sent. This event contains the list of scheduled callbacks (see [callbackInfo](#)).

If an administrator adds or removes an agent's callbacks, the [callbackAdded](#) or [callbackRemoved](#) event is sent. Each event contains a list of callbacks that have been added or removed.

To perform a callback, send a [makeCall](#) request that contains the ID of the callback. The CTI Web Services then remove the scheduled callback from the list.

addCallback

This method is no longer used. See `addCallback2` or `addCallback3`.

addCallback2

Use this method to schedule a callback associated with a campaign.

addCallback2.

Parameter	Type	Description
number	string	Callback number.
duedate	long	Reminder date in milliseconds (Epoch time).
comments	string	Callback comments.
campaignId	long	ID of the campaign with which to associate the callback.

emptyResponse.

addCallback3

Use this method to schedule a callback associated with a CRM record in a campaign. At the due date and time of the callback, the agent receives a message. The agent can perform the callback, delay it until later during the same session, dismiss it until the next session, or delete it.

addCallback3.

Parameter	Type	Description
number	string	Callback number.
duedate	long	Reminder date in milliseconds (Epoch time).
comments	string	Callback comments.
campaignId	long	ID of the campaign with which to associate the callback.

associateCRMRecord	boolean	Whether the callback is associated with a CRM record: The CTI Web Services match the phone number to a contact record that contains the phone number.
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[emptyResponse.](#)

answerLiveCallback

Use this method to accept the queue callback offered.

[answerLiveCallback.](#) This method contains no parameters.

[emptyResponse.](#)

checkAndFilterCallback

Use this method to check if the number entered for the callback is valid and to change the number from a string to an integer.

[checkAndFilterCallback.](#)

Parameter	Type	Description
number	string	Phone number as string, for example: (555) 123-4567.

[checkAndFilterCallbackResponse.](#)

Parameter	Type	Description
return	string	Phone number as integer, for example: 5551234567.

dismissCallback

Use this method to ignore the callback reminder during the current session. The agent is reminded after logging in the next time.

dismissCallback.

Parameter	Type	Description
id	long	Callback ID.

emptyResponse.

getCallbacks

Use this method to obtain a list of scheduled callbacks.

getCallbacks. This method contains no parameters.

getCallbacksResponse.

Parameter	Type	Description
return	callbackInfo [0..unbounded]	List of callbacks.

getCRMForCallbackId

Use this method to obtain the CRM ID for the callback.

getCRMForCallbackId.

Parameter	Type	Description
id	long	CRM ID for the callback.

getCRMForCallbackIdResponse.

Parameter	Type	Description
return	map	Key-value pairs that are the field names and values in the Five9 contact database.

getCRMMap

Use this method to obtain a mapping of the field ID to the field name.

getCRMap. This method contains no parameters.

getCRMapResponse.

Parameter	Type	Description
return	map	Key-value pairs that are the field names and values in the Five9 contact database.

rejectLiveCallback

Use this method to reject the queue callback currently offered.

rejectLiveCallback. This method contains no parameters.

emptyResponse.

Parameter	Type	Description
	int	Always 0.

removeCallback

Use this method to permanently delete a callback number.

removeCallback.

Parameter	Type	Description
id	long	ID of the callback number to remove.

emptyResponse.

Parameter	Type	Description
	int	Always 0.

resetRemindedCallback

Use this method to reset a callback.

resetRemindedCallback.

Parameter	Type	Description
id	long	Callback ID.

emptyResponse.

Parameter	Type	Description
	int	Always 0.

snoozeCallback

Callbacks reminders automatically send [callbackOverdue](#) events to inform the client about overdue callbacks. Use this method to ignore a callback reminder temporarily during the current session.

snoozeCallback.

Parameter	Type	Description
id	long	Callback ID.
time	long	Duration of snooze time.

emptyResponse.

Parameter	Type	Description
	int	Always 0.

updateCallback

Use this method to update a callback number.

updateCallback.

Parameter	Type	Description
info	callbackInfo	Updated value from getCallbacks .

emptyResponse.

Parameter	Type	Description
	int	Always 0.

Chat Sessions

Use these methods to manage chat sessions and users.

[acceptChatRequest](#)[createChat](#)[createChatSession](#)[endChat](#)[exitChat](#)[getLoggedUsers](#)[inviteUserToChat](#)[kickUserFromChat](#)[processChatInvite](#)[requestHelpChat](#)[sendBroadcastMessage](#)[sendChatMessage](#)[sendSkillBroadcastMessage](#)[sendUserMessage](#)

acceptChatRequest

Use this method to allow a user to accept a chat invitation after receiving a chatInvite event.

acceptChatRequest.

Parameter	Type	Description
chatId	string	Chat ID.

emptyResponse.

Parameter	Type	Description
		None.

createChat

Use this method to create a chat session.

createChat. This method contains no parameters.

createChatResponse.

Parameter	Type	Description
return	string	Chat ID.

createChatSession

Use this method to create a chat session.

createChatSession. This method contains no parameters.

emptyResponse.

endChat

Use this method to end a chat session.

endChat.

Parameter	Type	Description
chatId	string	Chat ID.

emptyResponse.

exitChat

Use this method to exit a chat session.

exitChat.

Parameter	Type	Description
chatId	string	Chat ID.

emptyResponse.

getLoggedUsers

Use this method to obtain a list of logged-in users.

getLoggedUsers. This method contains no parameters.

getLoggedUsersResponse.

Parameter	Type	Description
return	userInfo [0..unbounded]	List of users who are online and information about each user.

inviteUserToChat

Use this method to invite a user to chat.

inviteUserToChat.

Parameter	Type	Description
chatId	string	Chat ID.
userInfo	userInfo	User information.

emptyResponse.

kickUserFromChat

Use this method to remove a user from a chat session.

kickUserFromChat.

Parameter	Type	Description
chatId	string	Chat ID.
userInfo	userInfo	User information.

emptyResponse.

processChatInvite

Use this method to accept or decline an invitation to chat.

`processChatInvite.`

Parameter	Type	Description
chatId	string	Chat ID.
accept	boolean	Whether to accept or decline a chat invitation.

`emptyResponse.`

requestHelpChat

Use this method to request help from a supervisor.

`requestHelpChat.` This method contains no parameters.

`emptyResponse.`

sendChatMessage

Use this method to send a chat message.

`sendChatMessage.`

Parameter	Type	Description
chatId	string	Chat ID.
message	string	Message string.

`emptyResponse.`

sendBroadcastMessage

Use this method to broadcast a message.

sendBroadcastMessage.

Parameter	Type	Description
message	string	Message string.

emptyResponse.

sendSkillBroadcastMessage

Use this method to broadcast a message to users with a specific skill.

sendSkillBroadcastMessage.

Parameter	Type	Description
skill	userInfo	Skill recipient of the message.
message	string	Message string.

emptyResponse.

sendUserMessage

Use this method to send a message to a user.

sendUserMessage.

Parameter	Type	Description
user	userInfo	User ID.
message	string	Message string.

emptyResponse.

Connectors

A connector can open a browser to make a POST or GET request.

[requestConnectors](#)
[processConnector](#)

[openBrowser](#)

Connectors, which are configured by the administrator for each campaign, can be automatic or manual:

- Automatic connectors are triggered when a specific event is received.
- Manual connectors are available to the client to start when necessary.

requestConnectors

Use this method to get a list of manual connectors available for a campaign. For information about automatic connectors, see the [openURL](#) event.

[requestConnectors.](#)

Parameter	Type	Description
campaignId	long	Campaign ID.

[requestConnectorsResponse.](#)

Parameter	Type	Description
return	connectorValue [0..unbounded]	List of connectors.

processConnector

Use this method to perform the action associated with a connector.

[processConnector.](#)

Parameter	Type	Description
campaignId	long	Campaign ID.
connectorId	long	Connector ID.

[emptyResponse.](#)

openBrowser

Use this method to open the default browser at the specified URL if the client cannot open the URL,

openBrowser.

Parameter	Type	Description
url	string	Required URL.

emptyResponse.

Contact Records

These methods enable you to manage CRM contact records.

[addCrmRecord](#)
[getCrmRecord](#)
[getCrmRecords](#)

[lookupCrmRecords](#)
[removeCrmRecord](#)
[updateCrmRecord](#)

addCrmRecord

This method adds a CRM contact record to the database. The response contains the difference between the record saved and the record passed to the method, such as record ID and modification time.

addCrmRecord.

Parameter	Type	Description
crmRecord	crmRecord	Attributes of the record to add to the database.

addCrmRecordResponse.

Parameter	Type	Description
return	crmRecord	Attributes of the record added to the database.

getCrmRecord

This method returns the attributes of a CRM contact record.

[getCrmRecord.](#)

Parameter	Type	Description
id	long	ID of the record to find.

[getCrmRecordResponse.](#)

Parameter	Type	Description
return	crmRecord	Attributes of the record.

getCrmRecords

This method returns the attributes of a list of CRM contact record.

[getCrmRecords.](#)

Parameter	Type	Description
ids	long [0..unbounded]	List of IDs of records to find.

[getCrmRecordsResponse.](#)

Parameter	Type	Description
return	crmRecord [0..unbounded]	Attributes of the records.

lookupCrmRecords

This method finds CRM contact records by using number or criteria.

[lookupCrmRecords.](#)

Parameter	Type	Description
-----------	------	-------------

number	string	Searches by using the phone number. The search uses all numbers available for the contact record, such as number1, number2, and number3.
criteria	map	Search criteria other than numbers. Phone numbers are ignored.
lookupCursor	lookupCursor	Search cursor to include in subsequent searches.
limit	int	Maximum number of CRM records returned in each <code>crmRecordLookupResult</code> .

lookupCrmRecordsResponse.

Parameter	Type	Description
return	crmRecordLookupResult	Results of a search.

removeCrmRecord

This method deletes a CRM contact record.

removeCrmRecord.

Parameter	Type	Description
id	long	ID of the record to delete.

emptyResponse.

updateCrmRecord

This method updates a CRM contact record. The response contains the difference between the record passed in the request and the modified record.

updateCrmRecord.

Parameter	Type	Description
-----------	------	-------------

crmRecord	crmRecord	Attributes of the record to modify.
-----------	---------------------------	-------------------------------------

updateCrmRecordResponse.

Parameter	Type	Description
return	crmRecord	Attributes of the modified record.

CTI Web Services

These methods enable you to obtain basic information about the CTI Web Services.

[checkLocalTime](#)
[getCurrentWebserver](#)
[getOption](#)
[getStatus](#)
[getVersion](#)
[sendHttpServerMessage](#)

checkLocalTime

Use this method to check local time with server time.

checkLocalTime.

Parameter	Type	Description
initial	boolean	Whether the local time matches the server time.

checkLocalTimeResponse.

Parameter	Type	Description
return	checkLocalTimeRes ult	Result of the check.

getCurrentWebserver

Use this method to obtain the URL of the Five9 server currently hosting the CTI Web Services.

getCurrentWebserver. This method contains no parameters.

getCurrentWebserverResponse.

Parameter	Type	Description
return	string	URL of the Web server.

getOption

Use this method to obtain the value of an option of the CTI Web Services.

getOption.

Parameter	Type	Description
option	string	Option name.

getOptionResponse.

Parameter	Type	Description
return	string	Option value.

getStatus

Use this method to obtain the current status of an agent.

getStatus. This method contains no parameters.

getStatusResponse.

Parameter	Type	Description
return	agentBridgeStatus	Information about an agent.

getVersion

Use this method to obtain the version of the CTI Web Services components.

getVersion.

Parameter	Type	Description
componentName	string	Component name.

getVersionResponse.

Parameter	Type	Description
return	string	Component version.

sendHttpServerMessage

[Note spelling.] Use this method to send a message to be forwarded by the HTTP server.

sendHttpServerMessage.

Parameter	Type	Description
message	string	Message.

emptyResponse.

Dispositions

[finishCall](#)
[finishCall2](#)
[getCampaignDispositions](#)
[getCampaignDispositions2](#)
[getDispositionsInfo](#)

After a call is terminated by the remote party or by the agent (see [disconnectCall](#)), the agent must assign a disposition. Disposition codes are defined in the Five9 VCC and in your CRM system. The codes may be the same or different in both systems. The disposition code is assigned in Five9 and possibly in your CRM system. To assign disposition codes correctly, the client needs to be aware of the mapping differences

between Five9 and your CRM system. After the disposition is assigned, the agent is ready for another call.

finishCall

After the client selects a disposition, the disposition needs to be set in the VCC.

finishCall.

Parameter	Type	Description
dispositionId	long	ID of the disposition.
comments	string	Agent's notes added to the Five9 call log reports.

emptyResponse.

finishCall2

Use this method if the disposition enables agents to change the redial time configured by the administrator.

finishCall2.

Parameter	Type	Description
dispositionId	long	ID of the disposition.
comments	string	Agent's notes added to the Five9 call log reports.
timeout	long	Redial timeout in milliseconds since the end of the call.

emptyResponse.

getCampaignDispositions

Use this method to obtain a list of disposition codes associated with a campaign. To enforce a time-out when selecting a disposition, see [getCampaignDispositions2](#).

getCampaignDispositions.

Parameter	Type	Description
campaignId	long	ID of the campaign.

getCampaignDispositionsResponse.

Parameter	Type	Description
return	dispositionValue [0 . . unbounded]	List of available dispositions, including system dispositions.

getCampaignDispositions2

Use this method to obtain a list of disposition codes associated with a campaign. Alternately, the client may automatically assign the disposition based on CRM criteria or call-scripting results.

getCampaignDispositions2.

Parameter	Type	Description
campaignId	long	ID of the campaign.

getCampaignDispositions2Response.

Parameter	Type	Description
return	dispositionValue2 [0 . . unbounded]	List of available dispositions, including system dispositions, and a time-out parameter when selecting a disposition code.

getDispositionsInfo

Use this method to obtain information about the dispositions available for a conference call in the specified campaign.

getDispositionsInfo.

Parameter	Type	Description
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campaignId	long	ID of the campaign.
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getDispositionsInfoResponse.

Parameter	Type	Description
return	dispositionsInfo	Information about the available dispositions.

DNC List

Use this method to manage your DNC list.

addNumberToDnc

Use this method to add numbers to a do-not-call (DNC) list. If the agent does not have permission to do so, or if the number cannot be added to the list, the [failedAddingNumberToDNC](#) event is sent.

addNumberToDnc.

Parameter	Type	Description
number	string	Number to be added to the DNC list.

emptyResponse.

Events

This section describes the methods that you can use to make the CTI Web Services available for requests across the network.

[addListener](#)
[checkEvent](#)
[removeListener](#)

addListener

Use this method to create an event listener that checks for events.

addListener.

Parameter	Type	Description
name	string	Name of the client receiver. If you use this name multiple times during a Web Services session, you receive an exception. For example, to guarantee that the listener name is unique, append the date and time to it.

addListenerResponse.

Parameter	Type	Description
return	string	ID of the receiver (listenerId) that identifies the registered client.

checkEvent

Use this method to get events from the listener queue and to prevent that your agents are timed out if they are inactive for a specified duration. This method is typically called in a loop by a background thread that is started after `addListener` is called. It is used to receive CTI events. As a side benefit, the method keeps the session active until you call `removeListener`. The Web Services client uses long polling to check periodically whether events have occurred.

checkEvent.

Parameter	Type	Description
listenerId	string	Receiver's ID returned by <code>addListener</code> .
timeout	int	Amount of time of the waiting period (ms).

checkEventResponse. The CTI Web Services determine whether events have occurred by looking at the agent's queue:

- If no event occurs before the time specified in the request, a timeout message is returned.
- If an event has occurred, an [event](#) object provides the time of the event. Some events contain additional call information. For a list of events, see [Events](#).

Parameter	Type	Description
return	event	Time of the event.

removeListener

Use this method to remove the Web Services client as a listener when it no longer needs to receive events.

[removeListener.](#)

Parameter	Type	Description
listenerId	string	Receiver's ID.

[EmptyResponse.](#)

Greetings

This section contains the methods that you can use to manage greeting messages.

resetGreeting

Use this method to reset the current greeting or to upload a new one. You receive an exception if no greeting was previously set.

[resetGreeting.](#) This method contains no parameters.

[emptyResponse.](#)

uploadGreeting

Use this method to upload a greeting. You receive an exception if the uploaded audio file type is not supported by the audio codec.

uploadGreeting.

Parameter	Type	Description
arg0	string	Path of the WAV audio file.

emptyResponse.

Scripts

Use this method to obtain script information.

requestScript

Use this method to request the script configured for the campaign.

requestScript.

Parameter	Type	Description
campaignId	long	ID of the campaign.

requestScriptResponse.

Parameter	Type	Description
return	string	Temporary script file name.

Skill Prompts

This section contains the methods that you can use to manage prompts.

[cancelDownloadSkillPrompt](#)

[getSkillPrompts](#)

[pauseSkillPrompt](#)

[playSkillPrompt](#)
[startDownloadSkillPrompt](#)
[stopSkillPrompt](#)

cancelDownloadSkillPrompt

Use this method to stop downloading a prompt. When the download is stopped, the [skillPromptDownloadCancelled](#) event is sent.

cancelDownloadSkillPrompt. This method contains no parameters.

emptyResponse.

getSkillPrompts

Use this method to obtain a list of skill prompts.

getSkillPrompts. This method contains no parameters.

getSkillPromptsResponse.

Parameter	Type	Description
return	agentPromptValue [0..unbounded]	List of skill prompts.

pauseSkillPrompt

Use this method to pause a skill prompt.

pauseSkillPrompt. This method contains no parameters.

emptyResponse.

playSkillPrompt

Use this method to play a skill prompt to all participants during a call.

playSkillPrompt.

Parameter	Type	Description
id	long	Prompt ID.

emptyResponse.

startDownloadSkillPrompt

Use this method to download a skill prompt from the server to a local file system. The [skillPromptDownloaded](#) event is sent afterward.

startDownloadSkillPrompt.

Parameter	Type	Description
id	long	Prompt ID.

emptyResponse.

stopSkillPrompt

Use this method to stop playing a skill prompt.

stopSkillPrompt. This method contains no parameters.

emptyResponse.

Softphone Options

Use these methods to manage all softphone options.

getSoftphoneAutoHungup	sendDTMF
getSoftphoneDevices	setMute
getSoftphoneInputDevices	setOption
getSoftphoneOutputDevices	setSoftphoneAutoHungup
getSoftphoneMicrophoneAgc	setSoftphoneDevices
getSoftphoneMicrophoneLevel	setSoftphoneMicrophoneAgc
getSoftphoneSpeakerVolume	setSoftphoneMicrophoneLevel
isSoftphoneAnswerRequired	setSoftphoneSpeakerVolume
makeAnswerOnHungup	softphoneHangupPhone
restartStation	softphonePickupPhone

getSoftphoneAutoHungup

[Note spelling.] Use this method to check if the softphone is set to automatically hang up after a call. If this feature is enabled, agents cannot receive calls automatically.

getSoftphoneAutoHungup. This method contains no parameters.

getSoftphoneAutoHungupResponse.

Parameter	Type	Description
return	boolean	Whether the automatic hang-up feature is enabled.

getSoftphoneDevices

Use this method to obtain the list of available input devices in the computer and to select the ring device, speaker device, and microphone.

getSoftphoneDevices. This method contains no parameters.

getSoftphoneDevicesResponse.

Parameter	Type	Description
return	int [0 . . unbounded]	Array of size 3 with indexes of devices in this order: ring, speaker, and microphone.

getSoftphoneInputDevices

Use this method to obtain the names of the available input devices.

getSoftphoneInputDevices. This method contains no parameters.

getSoftphoneInputDevicesResponse.

Parameter	Type	Description
return	string [0..unbounded]	List of the available input devices.

getSoftphoneOutputDevices

Use this method to obtain the names of the available output devices.

getSoftphoneOutputDevices. This method contains no parameters.

getSoftphoneOutputDevicesResponse.

Parameter	Type	Description
return	string [0..unbounded]	List of the available output devices.

getSoftphoneMicrophoneAgc

Use this method to check whether the automatic gain of the microphone is on or off. The automatic gain is the ability to adjust the volume according to the strength of the electronic signal.

getSoftphoneMicrophoneAgc. This method contains no parameters.

getSoftphoneMicrophoneAgcResponse.

Parameter	Type	Description
return	boolean	Whether the automatic gain is on (default) or off.

getSoftphoneMicrophoneLevel

Use this method to obtain the microphone volume.

getSoftphoneMicrophoneLevel. This method contains no parameters.

getSoftphoneMicrophoneLevelResponse.

Parameter	Type	Description
return	int	Volume value from 0 to 100.

getSoftphoneSpeakerVolume

Use this method to obtain the speaker volume.

getSoftphoneSpeakerVolume. This method contains no parameters.

getSoftphoneSpeakerVolumeResponse.

Parameter	Type	Description
return	int	Volume value from 0 to 100.

isSoftphoneAnswerRequired

Use this method to find out whether the answer button is visible.

isSoftphoneAnswerRequired. This method contains no parameters.

isSoftphoneAnswerRequiredResponse.

Parameter	Type	Description
return	boolean	Whether the answer button is visible.

makeAnswerOnHungup

[Note spelling.] Use this method to answer a call after receiving the [answerCallIncoming](#) event. If automatic hang-up is enabled, the answerCallIncoming event is sent for each incoming call. When the event flag=true, additional interaction, such as the client adding an answer call button, may be needed in case of softphone hang up. If the client

does not pick up the call within a specific period of time, a second event is sent with `flag=false`, which means that pick up is no longer possible.

makeAnswerOnHungup. This method contains no parameters.

emptyResponse.

restartStation

Usually, if the station type is selected, the CTI Web Services restart the softphone station automatically after a successful login. If this does not occur, use this method to restart the station manually. To indicate the result of the restart, one of these events is sent: `restartStationDone` or [restartStationFailed](#), with a reason for the failure.

restartStation. This method contains no parameters.

emptyResponse.

sendDTMF

Use this method to enable a softphone to send DTMF tones.

sendDTMF.

Parameter	Type	Description
digit	int	Digit (0–9, or 10–‘*’ and 11–‘#’).

emptyResponse.

setMute

Use this method to mute the softphone on or off.

setMute.

Parameter	Type	Description
mute	boolean	Whether to mute the softphone.

emptyResponse.

setOption

Use this method to change sound and methods for receiving incoming calls. When phones are set to answer automatically, calls are active before agents receive the call, which prevents them from seeing the call information. To display the call information before the call becomes active, disable the [canConfigureAutoAnswer](#) permission so that you can set the calls to be answered a set number of seconds after the `incomingCall` event. If agents do not answer before the time expires, the calls are reassigned to other agents.

setOption.

Parameter	Type	Description
option	string	Whether to mute the softphone. • <code>alerts.callback_reminders</code>: Whether to play a sound for callback reminders. • <code>alerts.sound_for_call</code>: Whether to play a sound for incoming calls. • <code>alerts.sound_for_callback</code>: Whether to play a sound for callbacks. • <code>alerts.sound_for_vm</code>: Whether to play a sound for incoming voicemail. • <code>audio.auto_gain</code>: Whether automatic gain is enabled. • <code>audio.auto_gain_enabled</code>: Whether automatic gain control is enabled. • <code>audio.auto_hangup</code>: Whether to disconnect the softphone after each call. If enabled, an Answer button appears for each incoming call. • <code>audio.mic_device</code>: Microphone device. • <code>audio.mic_vol</code>: Microphone volume.

value	string	• <code>audio.ring_device</code>: Device used to play incoming calls, such as headset or the sound card. • <code>audio.speaker_device</code>: Primary source of sound. • <code>audio.speaker_vol</code>: Speaker volume setting. • <code>autoanswer.inbound</code>: Whether to answer inbound calls automatically. • <code>autoanswer.internal</code>: Whether to answer internal calls automatically. • <code>autoanswer.outbound</code>: Whether to answer outbound calls automatically. • <code>dialpad.before_connection</code>: Whether to show the dial pad before the connection is established. Read-only option. • <code>general.auto_login</code>: Whether the automatic login option is enabled. Disabled by default. • <code>general.launch_notify</code>: Whether to notify you after the CTI Web Services are started. Disabled by default. • <code>general.show_cavs</code>: Whether to show call variables to agents. • <code>javascript.loglevel</code>: Log level, for example: -1. • <code>login.password</code>: Masked password of the logged-in user. • <code>login.remember_me</code>: Whether the Remember Me option is enabled. • <code>login.station</code>: Station ID. • <code>login.station_type</code>: Type of station, for example: SOFTPHONE • <code>login.username</code>: Name of logged-in user. • <code>pbxMode</code>: Use the PBX mode.
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- `sfopencti.show_softphone_on_ringing`: Applies only to Salesforce Open CTI. Whether to show the softphone when the phone starts ringing.
- `tone.autodial.call`: Play a tone for incoming autodial calls.
- `tone.extension.call`: Play a tone for incoming internal calls.
- `tone.inbound.call`: Play a tone for incoming inbound calls.
- `tone.outbound.call`: Play a tone for incoming outbound calls.
- `tone.preview.call`: Play a tone for incoming preview calls.

`emptyResponse.`

setSoftphoneAutoHungup

[Note spelling.] Use this method to enable or disable the automatic hang-up feature.

`setSoftphoneAutoHungup.`

Parameter	Type	Description
<code>value</code>	<code>boolean</code>	True to enable, false otherwise.

`emptyResponse.`

setSoftphoneDevices

Use this method to set a new softphone configuration.

`setSoftphoneDevices.`

Parameter	Type	Description
<code>ringDevice</code>	<code>int</code>	Output device index.
<code>speakerDevice</code>	<code>int</code>	Output device index.

microphoneDevice	int	Input device index.
micArg	boolean	Whether to turn on or off the automatic gain of the microphone.

[emptyResponse.](#)

setSoftphoneMicrophoneAgc

Use this method to set the automatic gain of the microphone.

[setSoftphoneMicrophoneAgc.](#)

Parameter	Type	Description
agc	boolean	Whether to turn on or off the automatic gain of the microphone.

[emptyResponse.](#)

setSoftphoneMicrophoneLevel

Use this method to set the microphone volume.

[setSoftphoneMicrophoneLevel.](#)

Parameter	Type	Description
level	int	Volume value from 0 to 100.

[emptyResponse.](#)

setSoftphoneSpeakerVolume

Use this method to set the speaker volume.

[setSoftphoneSpeakerVolume.](#)

Parameter	Type	Description
volume	int	Volume value from 0 to 100.

[emptyResponse.](#)

softphoneHangupPhone

Use this method to hang up a softphone connection.

[softphoneHangupPhone.](#) This method contains no parameters.

[emptyResponse.](#)

softphonePickupPhone

Use this method to pick up a softphone connection.

[softphonePickupPhone.](#) This method contains no parameters.

[emptyResponse.](#)

User Account

This section describes the methods that you can use to act on a user account.

[cachedLogin](#)

[login](#)

[loginAsync](#)

[loginAsync2](#)

[loginAsync3](#)

[logout](#)

[logoutAndExit](#)

[changePassword](#)

[getSelfInfo](#)

[setActiveSkills](#)

[reconnect](#)

[runVoiceConnectivityTest](#)

cachedLogin

Use this method to retrieve a cached password.

[cachedLogin.](#)

Parameter	Type	Description
password	string	Agent's password.

[emptyResponse.](#)

login

Use this synchronous method if the client can wait before performing another operation.

After calling any of the login methods, the [loginProcessStarted](#) event is sent. The login process can also be initiated from another Web Services client. After the login process is completed successfully, the [logoutProcessFinished](#) event is sent.

Important Before logging users into a PSTN station or forwarding calls between sessions, be sure that these users have permission from the administrator because they may incur high long-distance fees.

[login.](#)

Parameter	Type	Description
userName	string	Agent's user name.
password	string	Agent's password.
stationId	string	Station ID (in case of PSTN, a phone number).
stationType	string	Possible values: • PSTN: PSTN connection. • SOFTPHONE: softphone connection. • VoIP: gateway connection.

[emptyResponse.](#)

loginAsync

This method is no longer in use. See `loginAsync2` and `loginAsync3`.

loginAsync2

Use an asynchronous method if the client needs to update the agent UI during the login process, which can take time, especially with a softphone connection.

Important Before logging users into a PSTN station or forwarding calls between sessions, be sure that these users have permission from the administrator because they may incur high long-distance fees.

loginAsync2.

Parameter	Type	Description
userName	string	Agent's user name.
password	string	Agent's password.
stationId	string	Station ID (in case of PSTN, a phone number).
stationType	string	Possible values: • PSTN: PSTN connection. • SOFTPHONE: softphone connection. • VoIP: gateway connection.
force	boolean	Whether to force the login.

emptyResponse.

loginAsync3

Use an asynchronous method if the client needs to update the agent UI during the login process, which can take time, especially with a softphone connection. This method is similar to loginAsync2, with the addition of the integrationType parameter.

Important Before logging users into a PSTN station or forwarding calls between sessions, be sure that these users have permission from the administrator because they may incur high long-distance fees.

loginAsync3.

Parameter	Type	Description
userName	string	Agent's user name.
password	string	Agent's password.
stationId	string	Station ID (in case of PSTN, a phone number).
stationType	string	Possible values: • PSTN: PSTN connection. • SOFTPHONE: softphone connection. • VoIP: gateway connection.
force	boolean	Whether to force the login.
integrationType	string	Type of integration.

emptyResponse.

logout

Use this method to log out an agent.

logout.

Parameter	Type	Description
reasonId	long	ID of the reason code.

emptyResponse.

logoutAndExit

Use this method to stop the CTI Web Services after logging out the agent. After the logout is completed, the `logoutProcessFinished` event is sent.

logoutAndExit.

Parameter	Type	Description
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reasonId	long	ID of the reason code.
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[emptyResponse.](#)

changePassword

Use this method to change the password after the agent has logged in.

[changePassword.](#)

Parameter	Type	Description
newPassword	string	Agent's new password.

[emptyResponse.](#)

getSelfInfo

Use this method to retrieve information about a user.

[getSelfInfo.](#) This method contains no parameters.

[getSelfInfoResponse.](#)

Parameter	Type	Description
return	userInfo	Information about a user.

setActiveSkills

If the user already has skills, use this method to activate the skills after the user has logged in successfully. The user cannot receive calls associated with a campaign until the skills are activated.

[setActiveSkills.](#)

Parameter	Type	Description
skills	long [0..unbounded]	List of skills.

[emptyResponse.](#)

reconnect

Use this method to reconnect a station.

[reconnect.](#)

Parameter	Type	Description
forced	boolean	Whether to force a new connection.

[emptyResponse.](#)

runVoiceConnectivityTest

Use this method to test the audio quality of the VoIP connection by making an echo call: the agent hears a prompt to speak into the microphone, the agent's speech is recorded by the system, and the recording is played to the agent.

[runVoiceConnectivityTest.](#) This method contains no parameters.

[emptyResponse.](#)

User Information

[checkCanCreateConferenceWithAddressType](#)
[checkCanMakeCallsTo](#)
[checkCanMakeCallsToAddressType](#)
[checkCanTransferToAddressType](#)
[getAddressType](#)
[getAgentInfo](#)
[getAgentPermissions](#)

[getAllSkills](#)
[getAvailableSkills](#)
[getCampaigns](#)
[getSkillGroups](#)
[getUserDetails](#)
[isAddressValid](#)
[updateReadyStateCommon](#)

checkCanCreateConferenceWithAddressType

Use this method to check if the agent has permission to create a conference with the specified addressType.

checkCanCreateConferenceWithAddressType.

Parameter	Type	Description
addressType	string	Possible recipients: • AGENT • SKILL • CAMPAIGN • EXTERNAL • SPEED_DIAL

checkCanCreateConferenceWithAddressTypeResponse.

Parameter	Type	Description
return	boolean	Whether the user can create a conference with the specified recipient.

checkCanMakeCallsTo

Use this method to find out if the agent has permission to make calls of specified type. To find out if the agent has permission to call speed dial numbers, use [checkCanMakeCallsToAddressType](#).

checkCanMakeCallsTo.

Parameter	Type	Description
type	int	Type of call: • 0: External number • 1: Agent • 2: Skill • 3: Call • 4: Campaign

checkCanMakeCallsToResponse.

Parameter	Type	Description
return	boolean	Whether the agent can make calls of the specified type. • True: Can make calls. • False: Cannot make calls.

checkCanMakeCallsToAddressType

Use this method to check if the agent has permission to call the specified addressType.

checkCanMakeCallsToAddressType.

Parameter	Type	Description
addressType	string	Possible recipients: • AGENT • SKILL • CAMPAIGN • EXTERNAL • SPEED_DIAL

checkCanMakeCallsToAddressTypeResponse.

Parameter	Type	Description
return	boolean	Whether the user can call the specified addressType.

checkCanTransferToAddressType

Use this method to check if the agent has permission to transfer a call to the specified addressType.

checkCanTransferToAddressType.

Parameter	Type	Description
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addressType	string	Possible recipients: • AGENT • SKILL • CAMPAIGN • EXTERNAL • SPEED_DIAL
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checkCanTransferToAddressTypeResponse.

Parameter	Type	Description
return	boolean	Whether the user can transfer a call to the specified addressType.

getAddressType

Use this method to obtain the type of call that corresponds to the phone number.

getAddressType.

Parameter	Type	Description
number	string	Phone number.

getAddressTypeResponse.

Parameter	Type	Description
return	int	Type of call recipient: • 0: Agent • 1: Skill • 2: Campaign • 3: External • 4: Speed_dial

getAgentInfo

Use this method to obtain the agent's name.

getAgentInfo.

Parameter	Type	Description
agentId	long	Agent ID.

getAgentInfoResponse.

Parameter	Type	Description
return	agentInfo	ID and skill name of agent.

getAgentPermissions

Use this method to obtain the permissions assigned to an agent.

getAgentPermissions. This method contains no parameters.

getAgentPermissionsResponse.

Parameter	Type	Description
return	agentPermission	Agent's permissions.

getAllSkills

Use this method to obtain the list of all the skills in the domain.

getAllSkills. This method contains no parameters.

getAllSkillsResponse.

Parameter	Type	Description
return	skillInfo [0..unbounded]	List of all skills.

getAvailableAgents

Use this method to obtain the list of agents in the domain.

getAvailableAgents. This method contains no parameters.

getAvailableAgentsResponse.

Parameter	Type	Description
return	agentInfo [0..unbounded]	List of all agents.

getAvailableSkills

Use this method to obtain the list of skills assigned to the logged-in agent.

getAvailableSkills. This method contains no parameters.

getAvailableSkillsResponse.

Parameter	Type	Description
return	skillInfo [0..unbounded]	List of all agent's skills.

getCampaigns

Use this method to obtain a list of campaigns available for the current user.

getCampaigns. This method contains no parameters.

getCampaignsResponse.

Parameter	Type	Description
return	campaignInfo [0..unbounded]	List of all campaigns.

getSkillGroups

Use this method to obtain the list of skill groups.

getSkillGroups. This method contains no parameters.

getSkillGroupsResponse.

Parameter	Type	Description
-----------	------	-------------

return	skillGroup [0..unbounded]	List of all skill groups.
--------	--	---------------------------

getUserDetails

Use this method to obtain additional details about the logged-in agent.

getUserDetails. This method contains no parameters.

getUserDetailsResponse.

Parameter	Type	Description
return	userDetails	Information about the logged-in agent.

isAddressValid

Use this method to verify whether a phone number is valid.

isAddressValid.

Parameter	Type	Description
number	string	Phone number to verify.

isAddressValidResponse.

Parameter	Type	Description
return	boolean	Whether the number is valid.

updateReadyStateCommon

Use this method to change the agent's state of readiness.

updateReadyStateCommon.

Parameter	Type	Description
state	readyStateValueCommon	List of states of readiness.

[emptyResponse.](#)

VCC Configuration

Use these methods to retrieve information from the Five9 Virtual Call Center.

[acknowledgeNotice](#)
[getAvailableCampaigns](#)
[getCampaignInfo](#)
[getCrmFieldDataTypes](#)
[getManualCallsConfig](#)
[getReasonCodes](#)

acknowledgeNotice

Use this method to acknowledge the VCC maintenance notice event that is sent if the administrator has configured the notice.

[acknowledgeNotice.](#)

Parameter	Type	Description
noticeid	long	Maintenance notice ID.
accept	boolean	Whether the notice is accepted.

[emptyResponse.](#)

getAvailableCampaigns

Use this method to obtain a list of campaigns available for the current user.

[getAvailableCampaigns.](#) This method contains no parameters.

[getAvailableCampaignsResponse.](#)

Parameter	Type	Description
return	campaignInfo [0..unbounded]	List of campaigns.

getCampaignInfo

Use this method to request the name of a campaign if the campaign ID was previously obtained from the IncomingCall event.

getCampaignInfo.

Parameter	Type	Description
campaignId	long	Campaign ID.

getCampaignInfoResponse.

Parameter	Type	Description
return	campaignInfo [0..unbounded]	Information about a campaign.

getCrmFieldDataTypes

Use this method to obtain the CRM field data types.

getCrmFieldDataTypes. This method contains no parameters.

getCrmFieldDataTypesResponse.

Parameter	Type	Description
return	crmFieldDataType [0..unbounded]	Array of CRM field data types.

getManualCallsConfig

Use this method to obtain the options for making manual calls.

getManualCallsConfig. This method contains no parameters.

getManualCallsConfigResponse.

Parameter	Type	Description
return	manualCallsConfig	Configuration for manual calls.

getReasonCodes

Use this method to request the list of logout reason codes.

getReasonCodes. This method contains no parameters.

getReasonCodesResponse.

Parameter	Type	Description
return	reasonCodeValue [0..unbounded]	List of logout reason codes.

isE164Enabled

Returns whether the domain is enabled for the E.164 list format.

isE164Enabled. Empty.

isE164EnabledResponse.

Parameter	Type	Description
return	boolean	Whether the domain is enabled.

Voicemail

This section contains the methods that you can use to manage personal and skill voicemail messages.

[Personal Recordings](#)

[Skill Messages](#)

Personal Recordings

This section describes how to manage personal voicemail recordings.

[playRecording](#)

[removeRecording](#)

[requestRecordings](#)

[saveRecording](#)

[stopPlayingRecording](#)

playRecording

Use this method to play a personal voicemail message.

playRecording.

Parameter	Type	Description
voicemailId	long	ID of the voicemail message.

emptyResponse.

removeRecording

Use this method to delete a personal voicemail recording.

removeRecording.

Parameter	Type	Description
voicemailId	long	ID of the voicemail message.

emptyResponse.

requestRecordings

Use this method to request a list of personal voicemail recordings after logging in. The client can also request the list of recordings at any time. The list of recordings is sent in the [recordingsChanged](#) event.

requestRecordings. This method contains no parameters.

emptyResponse.

saveRecording

Use this method to save a personal recording.

saveRecording.

Parameter	Type	Description
voicemailId	long	ID of the voicemail message.
path	string	File path of the saved recording.

emptyResponse.

stopPlayingRecording

Use this method to stop playing a personal recording. The [recordingPlayFinished](#) event is sent.

stopPlayingRecording. This method contains no parameters.

emptyResponse.

Skill Messages

Voicemail messages can be associated with a skill. When a skill routing module is used, you can offer to the caller the option to leave a voicemail message for a particular skill. These messages are then routed to agents who possess that skill.

[acceptVoicemail](#)
[deleteVoicemail](#)
[playVoicemail](#)
[processVoicemail](#)
[rejectVoicemail](#)
[saveVoicemail](#)
[transferVoicemail](#)

To receive and process skill voicemail messages, agents need to be in the Ready Call&VM or Ready VM state. When a skill voicemail message is delivered to an agent or to a client, the [voicemailDelivered](#) event is sent.

acceptVoicemail

Use this method to accept skill voicemail messages. When voicemail messages are presented, they must be accepted before the time-out set by the administrator. Otherwise, the VCC takes back the message.

acceptVoicemail. This method contains no parameters.

emptyResponse.

To confirm that a voicemail message was accepted, these events are sent:

- [voicemailAccepted](#): detailed information about the voicemail message.
- [voicemailCRMRecordChanged](#): information about the caller (if inbound call) or contact (if outbound call).

After a voicemail message is accepted, it can be played, deleted, processed, and transferred.

deleteVoicemail

Use this method to delete a skill voicemail message.

deleteVoicemail.

Parameter	Type	Description
comments	string	Comments associated with the voicemail message.

emptyResponse.

playVoicemail

Use this method to play a skill voicemail message.

playVoicemail. This method contains no parameters.

emptyResponse.

processVoicemail

Use this method to finish processing a skill voicemail message.

processVoicemail.

Parameter	Type	Description
comments	string	Comments associated with the voicemail message.

emptyResponse.

rejectVoicemail

Use this method to reject a skill voicemail message.

rejectVoicemail. This method contains no parameters.

emptyResponse.

saveVoicemail

Use this method to save a skill voicemail message as an audio file.

saveVoicemail.

Parameter	Type	Description
path	string	Path to the saved message.

emptyResponse.

transferVoicemail

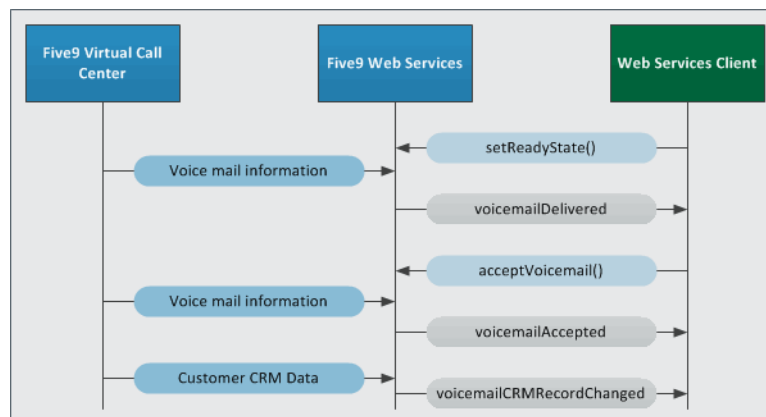
Use this method to transfer a voicemail message.

transferVoicemail.

Parameter	Type	Description
number	string	Number (same as for makeCall).

emptyResponse.

To confirm that the voicemail task is finished, the `voicemailEnded` event is sent. This figure shows the flow of events associated with receiving a voicemail message.



Worksheets

This section contains the methods that you can use to manage worksheets associated with campaigns.

[getWorksheetAnswers](#)
[hasWorksheet](#)
[isWorksheetComplete](#)

[requestWorksheet](#)
[saveWorksheetAnswers](#)
[updateWorksheetAnswers](#)

getWorksheetAnswers

Use this method to retrieve the worksheet answers for the current call after the answers have been saved. If the worksheet needs to be modified after it has been saved, use this method with [saveWorksheetAnswers](#).

getWorksheetAnswers. This method contains no parameters.

getWorksheetAnswersResponse.

Parameter	Type	Description
return	map	Name and values of worksheet answers.

hasWorksheet

Use this method to check if a worksheet exists for the campaign.

hasWorksheet.

Parameter	Type	Description
campaignId	long	ID of the campaign.

hasWorksheetResponse.

Parameter	Type	Description
return	boolean	Whether the campaign has a worksheet.

isWorksheetComplete

Use this method to check if a worksheet is completed.

isWorksheetComplete. This method contains no parameters.

isWorksheetCompleteResponse.

Parameter	Type	Description
return	boolean	Whether the worksheet is completed.

requestWorksheet

Use this method to request the worksheet of a campaign.

requestWorksheet.

Parameter	Type	Description
campaignId	long	ID of the campaign.

requestWorksheetResponse.

Parameter	Type	Description
return	questionValue [0..unbounded]	List of worksheet questions.

saveWorksheetAnswers

Use this method to save worksheet results after the agent has completed the worksheet.

saveWorksheetAnswers.

Parameter	Type	Description
values	map	Worksheet answers.

emptyResponse.

updateWorksheetAnswers

Use this method to update a single answer before the worksheet has been completed.

updateWorksheetAnswers.

Parameter	Type	Description
id	long	Question ID.
answer	string	Worksheet answer.

emptyResponse.

Chapter 4

Data Types

The data types specific to the Five9 methods are at http://localhost:8080/agent/agent_v2.types.xsd. For information on primitive data types, refer to www.w3.org/TR/xmlschema-2/.

agentBridgeStatus

This data type describes a agent's VCC session.

Name	Type	Description
callConnected	boolean	Whether the current call is connected.
currentCall	callInfo	Information about the current call.
loggedIn	boolean	Whether the agent is logged in.
onCall	boolean	Whether the agent on active call.
ready	boolean	Whether the agent is in Ready Call or Ready Call&VM states.
restarting	boolean	Whether the agent is restarting the station.

agentInfo

This data type describes the identity of an agent.

Name	Type	Description
agentId	long	Name of agent.
name	string	ID of agent.

agentPermission

This data type describes the possible permissions that can be assigned to an agent.

Name	Type	Description
canAddDNC	boolean	Can add numbers to the DNC list by using a system disposition.
canChangePassword	boolean	Can change password.
canCreateCallbacks	boolean	Can schedule a callback.
canCreateChat	boolean	Can initiate a chat session.
canCreateConference	boolean	Can create a conference.
canDeleteVoicemailSession	boolean	Can delete voicemail messages.
canDialDNC	boolean	Can make manual calls to the numbers listed in the DNC list. Without this permission, agents see an error message when dialing a number in the list.
canEditCRMSessions	boolean	Can edit contact data saved in the Five9 contact database.
canMakeCalls	boolean	Can initiate an outbound call.
canMakeInternalCalls	boolean	Can call another agent, supervisor, administrator, or skill.
canMakeRecordings	boolean	Can record greetings or prompts.
canMakeTestCalls	boolean	Can make test calls.
canManageSkillAvailability		Can select skill groups (ACD queues) when logging in.
canPlayPromptToCaller	boolean	Can play prerecorded prompts to caller.
canProcessVoicemail	boolean	Can process skill voicemail messages.
canRejectCalls	boolean	Can reject calls.
canRemoveCRMData	boolean	Can remove data from the contact database.
canSendMessages	boolean	Can send instant messages.

Name	Type	Description
canSkipNumbersInPreview	boolean	Can skip preview numbers.
canTransfer	boolean	Can transfer a call.
canTransferVoicemail	boolean	Can transfer a voicemail message.
canUseConnectors	boolean	Can use manual connectors.
reasonCodesEnabled	boolean	Can assign reason codes when selecting the Not Ready state or logging out.
canHold	boolean	Can place a call on hold.
canWrapUpCalls	boolean	Can end a call and assign a disposition afterward. When this permission is disabled, the agent can end a call only by assigning a disposition.
canPark	boolean	Can park a call.
canPickSalesforceObjectForCallLog	boolean	Applies only to Salesforce integrations. Can associate a call with a Salesforce object, such as an account, lead, or other object.
canTransferToAgents	boolean	Can transfer calls to agents.
canTransferToSkills	boolean	Can transfer calls to skill groups.
canCreateConferenceWithAgents	boolean	Can create conference calls with agents.
canCreateConferenceUsingSkills	boolean	Can create conference calls by using skill groups.

Name	Type	Description
canConfigureAutoAnswer	boolean	Can disable the auto-answer options in the Agent application, which forces agents to manually answer the calls by clicking Accept. Enabled by default. This permission does not affect the Hang-up Phone When Call Ended setting in the softphone options. Important For outbound calls placed by the dialer, instruct your agents to never disable automatic answer because the time allowed to answer the call expires before agents can answer.
canSelectSysRecycledDisposition	boolean	Can select a disposition that enables the agent to redial the number.
isTextEnabled	boolean	Can send and receive text channel messages.
chatMediaTypeEnabled	boolean	Can send and receive chat messages.
socialMediaTypeEnabled	boolean	Can send and receive social media messages.
emailMediaTypeEnabled	boolean	Can send and receive email messages.

agentPromptValue

This data type describes an agent prompt.

Name	Type	Description
description	string	Description of prompt.
description_is_set	boolean	Description of prompt is set.
duration	long	Duration of prompt.
duration_is_set	boolean	Duration of prompt is set.
id	long	ID of prompt.

Name	Type	Description
id_is_set	boolean	ID of prompt is set.
name	string	Name of prompt.
name_is_set	boolean	Name of prompt is set.

agentState

This data type describes summary information about the state of an agent.

Name	Type	Description
agentId	long	ID of the agent.
state	int	State of the agent.

agentStateValue

This data type describes detailed information about the state of an agent.

Name	Type	Description
agentId	long	ID of the agent.
state	string	State of the agent.
fullname	string	First and last names of the agent.
stateSince	string	Date and time of the status.
callType	int	If agent is on call, type of call. Otherwise -1. • 0: External number. • 1: Agent. • 2: Skill.
availableSkills	long [0..unbounded]	List of skills IDs assigned to the agent.

answerCriteriaValue

This data type describes an array of answer criteria.

Name	Type	Description
questionToJumpTo	string	Name of next question.
questionType	int	Question type: • 0: one line text • 1: text area • 2: check box • 3: combo box • 4: date • 5: information (no input required)
valueToCheck	string	Value for check box type.

callbackInfo

This data type describes the information about a callback.

Name	Type	Description
agent	long	Agent ID.
CRMId	long	ID of the CRM record.
campaignId	long	Campaign ID.
comments	string	Callback comments.
dueDate	long	Reminder date in milliseconds (Epoch time).
flags	long	16th bit: the callback was completed.
id	long	Callback ID used in makeCall.
number	string	Number to call back.
userGroup	long	User group ID.

callInfo

This data type describes the information about a call.

Name	Type	Description
agentInitiated	boolean	Whether the call was initiated by an agent, such as in internal call.
ani	string	Caller's phone number.
autoRecord	boolean	Whether the call is recorded. Administrators can record all calls made to an agent by setting an option in the agent's profile.
avgSessionTime	long	Average call time, which can be used to estimate call length.
callSessionId	string	Unique identifier for a call.
callType	int	0x0001: outbound 0x0002: inbound 0x0003: test 0x0004: agent call 0x0005: auto-dial call 0x0006: internal call 0x0007: queue callback 0x0101: inbound call voicemail 0x0102: outbound call voicemail 0x0103: internal voicemail from internal call
callbackId	long	Callback identifier.
campaignId	long	Campaign identifier.
comments	string	Comments previously entered in the call log by the transferring agent or when creating a callback.
consult	boolean	Whether...
dnis	string	Number dialed by caller.

Name	Type	Description
inboundPause	int	Pause before agent is connected to incoming call, for example when customer may be listening to a prompt.
internalReceiver	boolean	No longer used.
number	string	Phone number of remote party.
transfer	boolean	Whether the call was transferred.
userTransferringId	long	Numeric user ID of the transferring agent.

campaignInfo

This data type describes a campaign.

Name	Type	Description
id	long	Campaign identifier.
name	string	Campaign name.
flags	long	
state	int	
mode	int	Dialing, start, or mode.
extension	string	Unique 4-digit number to be used as the campaign's internal extension.

cavFullValue

This data type describes a call-attached variable.

Name	Type	Description
applyToAllDispositions	boolean	Whether to apply the variable to all dispositions.

Name	Type	Description
CCF	boolean	True if the variable is a custom call field, false if it is a system call field.
cavGroupId	long	Group ID of the variable.
defaultValue	string	Default value of the variable.
dscription	string	Description of the variable.
groupName	string	Group name of the variable.
id	long	ID of the variable.
name	string	Name of the variable.
systemId	long	System ID of the variable.
type	long	Type ID of the variable.
value	string	Value of the variable.

cavViewFullValue

This data type describes a call-attached variable.

Name	Type	Description
cavId	long	ID of the variable.
editable	boolean	Whether the variable can be edited.
id	long	ID of the variable.
name	string	Name of the variable.
order	int	
type	int	
width	int	

checkLocalTimeResult

This data type describes the times.

Name	Type	Description
ok	boolean	
localTime	long	Local time.
serverTime	long	Server time.

connectorValue

This data type describes a connector.

Name	Type	Description
className	string	Connector class.
description	string	Connector description.
dispositionIds	long [0..unbounded]	List of dispositions.
flags	long	Connector flags.
id	long	Connector ID.
name	string	Connector name.
url	string	URL of connector.

crmFieldType

This data type describes the restrictions of data types.

Name	Type	Description
id	long	CRM field data type ID.
name	string	CRM field data type name.

Name	Type	Description
restrictions	crmFieldRestrictionType [0..unbounded]	Array of CRM field restrictions.

crmFieldRestrictionType

This data type describes a restriction value.

Name	Type	Description
optional	boolean	Whether restriction is optional.
value	string	Restriction value.

crmRecord

This data type describes the attributes of a contact record.

Name	Type	Description
attributes	mapValue [0..unbounded]	Key and value details of the mapping.
recordID	long	ID of the contact record.
modificationTime	long	Date and time of the last record modification.
creationTime	long	Date and time of record creation.
numberDncStatus	boolean [0..unbounded]	Whether each of the phone number in the record is on your DNC list. • True: Specified number is on the DNC list. • False: Specified number is not on the DNC list.

crmRecordLookupResult

This data type describes the results of a search.

Name	Type	Description
crmRecords	crmRecord [0..unbounded]	List of contact records returned by the search.
hasMoreCrmRecords	boolean	Whether more crmRecords that meet the search criteria exist. • True: More crmRecords exist. The search can be continued. See lookupCursor below. • False: No more crmRecords exist. The search has ended.
lookupCursor	lookupCursor	Search cursor to include in the subsequent search (lookupCrmRecords).

dispositionsInfo

This data type describes disposition values.

Name	Type	Description
conferenceDispositio ning	long	
dispositionValue	dispositionValue2 [0..unbounded]	Information about the disposition values.
participants	participantInfo [0..unbounded]	List of participants.

dispositionValue

This data type describes a disposition value without time-out.

Name	Type	Description
description	string	Description of the disposition value.
flags	long	Bit number and description: • 0: first bit. • 16: Send mail notification. • 17: Send IM notification. • 18: Agent must confirm. • 19: Redial. • 20: Adds numbers to DNC list. • 21: Agent must complete worksheet. • 22: Track as first call resolution. • 23: Do not dial this number for campaign (DND disposition). • 24: Use timer to reactivate number (for DND disposition). • 25: Use redial timer (for redial disposition). • 26: Apply to Campaigns using this Disposition (for final disposition). • 27: Cannot be used in IVR. • 28: Allow agent to reactivation time (for DND disposition). • 29: Allow agent to change redial time (for redial disposition). • 30: Add active number and finalize record for campaign (for DNC disposition). • 31: Add all numbers and finalize record for campaign (for DNC disposition). • 32: Reset attempts counter.

Name	Type	Description
		The disposition <i>Add active number</i> (for DNC dispositions) if bits 30 and 31 are equals to zero. The disposition is final for CRM contacts if bits 19, 20, 23 are equals to zero. Other flags, which are not used, are equal to zero. Most flags can be set also in the administrator's VCC desktop.
id	long	ID of the disposition value.
name	string	Name of the disposition.

dispositionValue2

This data type describes a disposition value with time-out.

Name	Type	Description
description	string	Description of the disposition value.
flags	long	Disposition flags listed in dispositionValue .
id	long	ID of the disposition value.
name	string	Name of the disposition.
timeout	long	Disposition timeout.

event

This data type provides the time of the event.

Name	Type	Description
timestamp	string	Time of the event.

liveCallbackInfo

This data type describes information about a callback.

Name	Type	Description
callSessionId	string	ID of the session.
campaignId	long	ID of the campaign.
number	string	Agent's number.
callbackNumber	string	Number to call back.
offerId	long	
skillId	long	ID of the skill.
transferModule	string	

loginInfo

This data type describes information about a logged-in user.

Name	Type	Description
Password	string	User's password.
stationId	string	User's station ID.
stationtype	string	Type of station.
username	string	User's name.

lookupCursor

For internal use only. This data type provides the search cursor for subsequent searches.

Name	Type	Description
data	string	

manualCallsConfig

This data type describes the configuration of manual calls.

Name	Type	Description
defCampaign	long	Campaign ID.
maySelect	boolean	Whether agent may change campaign association.
maySelectNone	boolean	Whether agent may select None as campaign association.

map

This data type describes the values of a mapping.

Name	Type	Description
attributes	mapValue [0..unbounded]	Value of the mapping.

mapValue

This data type describes the details of a mapping, which are the field names and values in the Five9 contact database.

Name	Type	Description
key	string	Answer ID.
value	string	Content of the answer.

numberRestrictionValue

This data type describes the dialing restriction of a number.

Name	Type	Description
dialrestricted	boolean	Whether a number is restricted.
number	int	Phone number.

parkedCallInfo

This data type describes a parked call.

Name	Type	Description
address	string	
addressType	int	
callSessionId	string	ID of the call session.
campaignId	long	ID of the campaign.
number	string	
worksheetFinished	boolean	Whether the worksheet is finished.

participantInfo

This data type provides information about a participant.

Name	Type	Description
callSessionId	string	ID of the call session.
campaignId	long	ID of the campaign.
customerTitle	string	Title of customer.

passwordPolicy

This data type describes the password requirements.

Name	Type	Description
aspect	string	
count	long	Number of required characters.
description	string	

questionRestriction

This data type describes restrictions associated with worksheet questions

Name	Type	Description
id	long	
optional	boolean	Whether the restriction is optional.
restrictionType	int	Type of restriction.
value	string	Value of the restriction.

questionValue

This data type describes the questions on a worksheet.

Name	Type	Description
answerCriteria	answerCriteriaValue [..unbounded]	Array of answer criteria.
dataViewType	int	Question dataViewType ID.
description	string	Description of the question.
fieldViewType	int	Question fieldViewType ID.
id	long	Question ID.
idx	int	Question index.

Name	Type	Description
name	string	Question title.
options	mapValue [..unbounded]	Array of answer options.
restrictions	questionRestriction [..unbounded]	Array of question restrictions.
type	int	Question type: • 0: one line text • 1: text area • 2: check box • 3: combo box • 4: date • 5: information (no input required)

queuedCall

This table describes the calls in a queue.

Name	Type	Description
callSessionId	string	ID of the call session.
campaignId	long	ID of the campaign.
created	long	Date and time when the queue was created.
customer	string	Name of customer.
personal	boolean	Whether the call is personal.
priority	int	Priority of the call.

readyStateValue

This table describes the ready state.

Name	Type	Description
id	long	Name of the ready state, which can be shown to the user.
name	string	Ready state ID.

readyStateValueCommon

This data type describes whether the agent is enabled for calls, voicemail, and text channels.

Name	Type	Description
call	boolean	Whether the agent is enabled for phone calls.
voicemail	boolean	Whether the agent is enabled for voicemail.
text	boolean	Whether the agent is enabled for text channels.

reasonCodeValue

This data type describes a reason code.

Name	Type	Description
id	long	ID of the reason code.
name	string	Name of the Not Ready reason code that can be shown to the user.

recordingValue

This data type describes a recorded call.

Name	Type	Description
callSessionId	string	ID of the call.
campaignId	long	Campaign ID.
created	long	Date created.
filename	string	File name specified if recording loaded.
flags	long	Recording flags. New recording = 18th bit.
id	long	Recording ID used when playing, stopping, or deleting the recording.
length	long	Length of recording.
comment	string	Notes about the recording.
name	string	Name of the recording.
number	string	Remote number.

skillInfo

This data type describes a skill.

Name	Type	Description
name	string	Skill name.
skillId	long	Skill ID.

skillGroup

This data type describes a skill group.

Name	Type	Description
skillId	long	Skill ID.
userIds	long [0..unbounded]	IDs of the users in the group.

speedDialInfo

This data type describes a speed dial entry.

Name	Type	Description
name	string	Speed dial identifier used to make call.
description	string	Speed dial description.
id	long	Speed dial ID.
number	string	Number associated with speed dial.

userDetails

This data type describes details about a user.

Name	Type	Description
domain	string	Domain name.
domainId	long	Domain ID.
extension	string	Extension.
flags	long	User flags.
fullName	string	First and last names.
station	string	Station number.

Name	Type	Description
stationType	string	Station type.
username	string	Account name.
sccUrl	string	Applies only to the Oracle Service Cloud adapter. Agents do not see this field, which is set in the add-in settings: SccWebSocketURL is the URL of the text channel server.
sccPassword	string	Applies only to the Oracle Service Cloud adapter. Agents do not see this field, which is set in the add-in settings: sccPassword is taken from the configuration file when UseConfigForSccLogin is enabled. It is used to manually connect to the text channel server.

userInfo

This data type describes details about a user.

Name	Type	Description
id	long	ID of the user.
name	string	User name.
role	string	Role of the user.
uniqueId	string	

Chapter 5

Events

Events are methods that notify you when something occurs. Events are located at `http://localhost:8080/agent/agent_v2.events.xsd`.

The base event is `event`. All other events extend `event`.

activeSkillsChanged

This event describes the changed active skills.

Name	Type	Description
activeSkills	long [0..unbounded]	List of changed active skills.

answerCallIncoming

This event indicates a new call.

Name	Type	Description
flag	boolean	Whether a new call is arriving.

bridgeConnected

This event, which contains no parameters, indicates an active connection to the VCC.

bridgeDisconnected

This event, which contains no parameters, indicates no connection to the VCC.

bridgeForcedLogout

This event, which contains no parameters, indicates that the user was logged out forcibly from the VCC.

callAbandoned

This event indicates that the call was abandoned, for example when a campaign is stopped forcibly.

Name	Type	Description
callSessionId	string	Unique call identifier.

callbackAdded

This event indicates that a new callback was saved.

Name	Type	Description
info	callbackInfo	Information about a callback.

callbackOverdue

This event indicates that a callback is overdue. If this event occurs while agents are making outbound calls, you have two options:

- You can notify your agents of the overdue callback without taking further action. When the agents have time, they can process the overdue callback.
- You can place the agents in an unavailable state and send a `makeCall` request.

Name	Type	Description
id	long	Callback ID.
time	long	Due time of callback.

callbackOverdueCleanup

This event, which contains no parameters, indicates that an overdue callback was removed.

callbackRemoved

This event indicates that a callback was removed.

Name	Type	Description
id	long	Callback ID.

callbacksChanged

This event indicates that the list of outstanding scheduled callbacks has been updated.

Name	Type	Description
info	callbackInfo [0..unbounded]	Information about callbacks.
operation	int	

callCRMRecordChanged

This event indicates that a CRM record is available for the current call.

Name	Type	Description
callSessionId	string	Unique call identifier.
record	map	

callEnded

This event indicates that a call has ended, but a disposition has not yet been assigned.

Name	Type	Description
callSessionId	string	Unique call identifier.
party	int	

callEndedByACW

This event indicates that the disposition was not assigned before the after call work (ACW) time-out.

Name	Type	Description
callSessionId	string	Unique call identifier.
dispositionId	long	ID of the disposition value.
timeout	long	After call work (ACW) time-out value.

callError

This event indicates that an error occurred after the call started.

Name	Type	Description
callSessionId	string	Unique call identifier.
cause	string	Cause of the error.
operation	string	Operation that caused the error.

callFinished

This event indicates that a call has ended, and a disposition has been assigned.

Name	Type	Description
callHandleTime	long	
callLength	long	Call duration.
callSessionId	string	Unique call identifier.
callStarted	long	
callWrapupTime	long	

callInitiated

This event indicates that a call was initiated, for example by `makeCall`.

Name	Type	Description
number	string	
type	int	

callOffHold

This event indicates that a call is no longer on hold.

Name	Type	Description
callSessionId	string	Unique call identifier.

callOnHold

This event indicates that a call was placed on hold.

Name	Type	Description
callSessionId	string	Unique call identifier.

callParked

This event indicates that a call was parked automatically.

Name	Type	Description
callSessionId	string	Unique call identifier.

callReparked

This event indicates that a call was parked again.

Name	Type	Description
callinfo	callInfo	Information about a call.

callStarted

This event indicates that a call was started.

Name	Type	Description
callSessionId	string	Unique call identifier.

callUpdated

This event indicates that a call was updated.

Name	Type	Description
callinfo	callInfo	Information about a call.

callWrapupTimerStarted

This event indicates that the timer for after call work (ACW) time has started.

Name	Type	Description
callSessionId	string	Unique call identifier.
timeout	long	After call work (ACW) time-out value.

callsQueueUpdated

This event indicates that a queue was updated.

Name	Type	Description
calls	queuedCall [0..unbounded]	Information about the calls in the queue.

campaignUpdate

This event indicates that campaign data was updated.

Name	Type	Description
campaignId	long	Campaign ID.

chatBroadcastMessageReceived

This event indicates that a message was received from the specified user.

Name	Type	Description
from	string	Sender of the message.
type	string	Type of message.
message	string	Content of the message.

chatEnded

This event indicates that a chat session has ended.

Name	Type	Description
chatId	string	Chat ID.

chatHelpRequested

This event indicates that a user requested help.

Name	Type	Description
who	userInfo	Details about the user.

chatInvite

This event indicates an invitation to chat.

Name	Type	Description
chatId	string	Chat ID.
request	boolean	Whether an invitation was sent.
requestor	userInfo	Details about the user who sent the invitation.

chatInviteCancelled

This event indicates that an invitation to chat was canceled.

Name	Type	Description
chatId	string	Chat ID.

chatKickedFrom

This event indicates that a user was removed from a chat session.

Name	Type	Description
chatId	string	Chat ID.
userInfo	userInfo	Details about the user.

chatKilled

This event indicates that a chat session was terminated.

Name	Type	Description
chatId	string	Chat ID.

Name	Type	Description
userInfo	userInfo	Details about the user.

chatLoggedUsersUpdated

This event indicates an updated list of the users participating in a chat session.

Name	Type	Description
chatId	string	Chat ID.
users	userInfo [0..unbounded]	Details about the user.

chatMessage

This event provides information about a chat message.

Name	Type	Description
message	string	Content of the message.
senderId	long	ID of the sender.
senderRole	string	Role of the sender.

chatMessageDialog

This event provides information about a message received from a user.

Name	Type	Description
from	string	Sender of the message.
msg	string	Content of the message.
sent	string	Recipient of the message.

Name	Type	Description
type	string	

chatMessageReceived

This event provides information about a chat message.

Name	Type	Description
chatId	string	Chat ID.
from	userInfo	Sender of the message.
message	string	Content of the chat message.

chatParticipantStarted

This event indicates that a participant was added to the chat session.

Name	Type	Description
chatId	string	Chat ID.
requestor	userInfo	Information about the user who added the participant.

chatParticipantUpdated

This event provides an updated list of participants.

Name	Type	Description
chatId	string	Chat ID.
participants	userInfo [0..unbounded]	List of participants.

chatUserDeclined

This event indicates that a user declined a chat invitation.

Name	Type	Description
chatId	string	Chat ID.
userInfo	userInfo	Information about the user who declined the invitation.

chatUserDidNotRespond

This event indicates that a user ignored a chat invitation.

Name	Type	Description
chatId	string	Chat ID.
userInfo	userInfo	Information about the user who ignored the invitation.

chatUserJoined

This event indicates that a user joined a chat session.

Name	Type	Description
chatId	string	Chat ID.
userInfo	userInfo	Information about the user who joined the chat session.

chatUserLeft

This event indicates that a user left a chat session.

Name	Type	Description
chatId	string	Chat ID.
userInfo	userInfo	Information about the user who left the chat session.

click2DialInitiated

This event indicates that a call was started by the `startClick2Dial` method.

Name	Type	Description
number	string	Number called.

conferenceCompleted

This event indicates that a conference was completed with the `completeConference` method.

Name	Type	Description
callSessionId	string	Unique call identifier.

conferenceCreateFailed

This event indicates that an error occurred during the creation of a conference.

Name	Type	Description
callSessionId	string	Unique call identifier.
reason	string	Cause of the error

Name	Type	Description
errorCode	int	Error code.

conferenceError

This event indicates that an error occurred during a conference.

Name	Type	Description
callSessionId	string	Unique call identifier.
cause	string	Cause of the error
operation	string	Process that caused the error.

conferenceParticipantAdded

This event indicates that a participant was added to a conference.

Name	Type	Description
callSessionId	string	Unique call identifier.
info	participantInfo	Information about the participant.

conferenceParticipantRemoved

This event indicates that a participant was removed from a conference.

Name	Type	Description
callSessionId	string	Unique call identifier.
info	participantInfo	Information about the participant.

consultCallDisconnected

This event indicates that a third-party call was disconnected during warm transfer.

Name	Type	Description
callSessionId	string	Unique call identifier.

endOfPreviewingFailed

This event indicates that an error occurred at the end of a call preview.

Name	Type	Description
callSessionId	string	Unique call identifier.

event

The base event provides the time of the event.

Name	Type	Description
timestamp	string	Date and time in this format: as yyyy-mm-dd hh:mm:ss, in which hh is in the 24-hour format.

executeAgentCallback

This event indicates that an agent performed a callback.

Name	Type	Description
data	string	Content of the callback.

failedAddingNumberToDNC

This event indicates that an error occurred while adding a number to DNC.

Name	Type	Description
cause	string	Cause of the error.
number	string	Phone number to be added to the DNC list.

forceSelfLogout

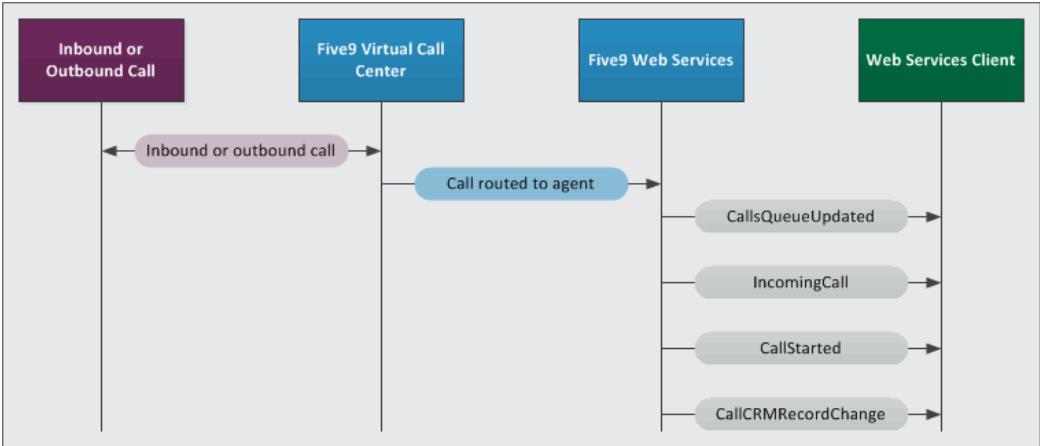
This event indicates that a user logged out of the VCC.

Name	Type	Description
loginInfo	loginInfo	Information about the user.

incomingCall

This event indicates a new call. Incoming call originate from callers or the Five9 dialer. This event contains the information that the client displays to the agent. This figure shows the flow of incoming call events.

Before agents receive a call, if you want the phone to ring long enough so that call information is displayed to agents before the call is active, modify the auto-answer settings of the phones. If the phones are set to answer calls automatically, the calls are active almost immediately. With the API, you can disable the auto-answer settings so that calls are answered a specified number of seconds after the incomingCall event. However, if agents do not answer before the time expires, the call is reassigned to another agent.



Name	Type	Description
callInfo	callInfo	Information about the call.

incomingCallInfo

This event indicates a new call. When automatic answer is disabled, this event notifies user of a new call. Follow up with the `makeCallAnswer` or `rejectCall` method.

Name	Type	Description
callInfo	callInfo	Information about the call.

invalidVersionEvent

This event, which contains no parameters, indicates that an error occurred when a user logged into the VCC.

liveCallbackEnded

This event indicates that a callback has ended.

Name	Type	Description
callSessionId	string	Unique call identifier.
offerId	long	

liveCallbackInitiating

This event indicates that a callback has started.

Name	Type	Description
info	liveCallbackInfo	Information about a callback.
autoAnswer	boolean	Whether auto-answer is enabled.

liveCallbackInitiatingFailed

This event indicates that a callback that was started has failed.

Name	Type	Description
callSessionId	string	Unique call identifier.
offerId	long	

liveCallbackCRMRecordChanged

This event indicates that a CRM record was changed after a callback.

Name	Type	Description
callSessionId	string	Unique call identifier.

Name	Type	Description
offerId	long	
record	map	Information about a mapping.

loginProcessFinished

This event indicates that the login process is completed.

Name	Type	Description
message	string	
stationNumber	string	User's station number
stationType	string	Type of station: softphone, PSTN, or gateway.
successfull	boolean	Whether the login process was successful.
user	string	Logged-in user.

loginProcessStarted

This event indicates that the login process has started.

Name	Type	Description
user	string	User information.

loginUIMessage

This event indicates that a login message was sent. The login message could contain a notification about a password change or password expiration.

Name	Type	Description
message	string	Login message.

logoutProcessFinished

This event, which contains no parameters, indicates that the logout process is completed.

maintenanceAnounce

[Note spelling.] This event indicates that the maintenance mode was announced.

Name	Type	Description
when	int	

maintenanceCanceled

This event, which contains no parameters, indicates that the maintenance mode was canceled.

maintenanceCompleted

This event, which contains no parameters, indicates that the maintenance mode was completed.

maintenanceNotice

This event describes a maintenance notice that appears immediately after logging into the VCC before work is started.

Name	Type	Description
annotation	string	Title of the notice.
id	long	ID of the notice
text	string	Content of the notice.

maintenanceRelogin

This event indicates whether logging in again is required.

Name	Type	Description
force	boolean	Whether to force users to log in again after maintenance.

maintenanceStarted

This event, which contains no parameters, indicates that the maintenance mode has started.

maintenanceStateChanged

This event indicates that the maintenance state has changed.

Name	Type	Description
hostType	int	
remainingTime	int	

makeCallFailed

This event indicates that an error occurred during a call.

Name	Type	Description
campaignId	long	Campaign ID.
cause	string	Cause of the error.
errorCode	int	Error code.
number	string	Phone number affected.
callbackId	long	Callback ID.

modelChanged

For internal use only.

Name	Type	Description
jsonData	string	Information about the model.

openURL

This event indicates that an a connector has been opened automatically. For information about manual connectors, see the [requestConnectors](#) method.

to open a URL (in triggers).

Name	Type	Description
newTab	boolean	Whether to open the URL in a new tab.
url	string	Connector URL to be opened and displayed to the agent.

optionsUpdated

This event indicates that options were updated.

Name	Type	Description
options	map	Value of a mapping

parkedCallDisconnected

This event indicates that a parked call was disconnected.

Name	Type	Description
callSessionId	string	Unique call identifier.

parkedCallsChanged

This event indicates that a list of parked calls has changed.

Name	Type	Description
parkedCalls	parkedCallInfo [0..unbounded]	Information about a list of parked calls.

parkError

This event indicates that an error occurred while a call was parked.

Name	Type	Description
callSessionId	string	Unique call identifier.
cause	string	Cause of the error.
operation	string	

passwordChangeRequired

This event indicates that a password must be changed.

Name	Type	Description
cause	string	Reason for changing the password.
mandatory <i>Note spelling.</i>	boolean	Whether the change is required.
policy	passwordPolicy [0..unbounded]	Information about passwords.

permissionsUpdated

This event indicates that user permissions were changed.

Name	Type	Description
permissions	agentPermission	Possible permissions.

previewCallInitiated

This event indicates that a call preview was dialed.

Name	Type	Description
callSessionId	string	Unique call identifier.

previewCRMRecordChanged

This event indicates that a CRM record is available for preview.

Name	Type	Description
callSessionId	string	Unique call identifier.

Name	Type	Description
record	map	
restriction	numberRestrictionValue [0..unbounded]	

previewDialingStarted

This event indicates that one of preview numbers was called.

Name	Type	Description
callinfo	callInfo	Call information.

previewEnded

This event indicates that a preview call has ended.

Name	Type	Description
callSessionId	string	Unique call identifier.
party	int	

previewInitiated

This event indicates that a preview call has started.

Name	Type	Description
callinfo	callInfo	Call information.

previewRenewed

This event indicates that a preview call was renewed.

Name	Type	Description
callSessionId	string	Unique call identifier.

previewWaitDisposition

This event indicates that a preview call requires a disposition. This event appears when a non-final disposition is selected for a preview number.

Name	Type	Description
callSessionId	string	Unique call identifier.
dispositionId	long	Disposition ID.

readyStateChanged

This event indicates that an agent's ready state was changed successfully.

Name	Type	Description
forced	boolean	Whether the change was forced.
id	long	ID of the change.
reasonCodeId	long	ID of the reason code.
whoForced	int	

readyStateChangedCommon

This event indicates that an agent was enabled for calls, voicemail, and text channels.

Name	Type	Description
readyStateCommon	readyStateValueCommon	List of states of readiness.

readyStatesUpdated

This event indicates that a list of ready states is available. This event is sent after the agent has logged in and has changed the state.

Name	Type	Description
states	readyStateValue [0..unbounded]	Information about the states.

reasonCodesEnabled

This event indicates whether reason codes are enabled.

Name	Type	Description
enabled	boolean	Whether reason codes are enabled.

recordingAdded

This event indicates that a personal voicemail recording was added.

Name	Type	Description
recording	recordingValue	Information about the recording.

recordingPlayFinished

This event indicates that a recorded personal voicemail was played.

Name	Type	Description
duration	long	Duration of the recording.

recordingStarted

This event indicates that a personal voicemail recording was started.

Name	Type	Description
callSessionId	string	Unique call identifier.
autoRecording	boolean	

recordingStopped

This event indicates that a personal voicemail recording was stopped.

Name	Type	Description
callSessionId	string	Unique call identifier.

recordingsChanged

This event indicates that a personal voicemail recording was changed.

Name	Type	Description
recordings	recordingValue [0..unbounded]	Information about the recording.

recordRestrictionsChanged

This event indicates that preview call restrictions were changed.

Name	Type	Description
callSessionId	string	Unique call identifier.
restriction	numberRestrictionValue [0..unbounded]	Information about dialing restrictions.

restartStationDone

This event, which contains no parameters, indicates that the station was restarted successfully.

restartStationFailed

This event indicates that an error occurred while the station was being restarted.

Name	Type	Description
cause	string	Cause of the failure.

restartStationStarted

This event, which contains no parameters, indicates that the station is being restarted.

setSoftphoneStatus

This event indicates that the softphone status was set.

Name	Type	Description
status	string	Current state of the softphone.

skillAdded

This event indicates that a skill was added.

Name	Type	Description
name	string	Name of skill.

skillPromptDownloadCancelled

This event, which contains no parameters, indicates that a skill prompt download was canceled.

skillPromptDownloadFailed

This event indicates that a skill prompt download failed.

Name	Type	Description
details	string	Information about the failure.

skillPromptDownloadProgress

This event indicates the progress of a skill prompt download.

Name	Type	Description
progress	double	Information about the download.

skillPromptDownloaded

This event indicates that a skill prompt was downloaded.

Name	Type	Description
filename	string	Name of the skill prompt file.

skillPromptPauseFailed

This event indicates that a skill prompt pause failed.

Name	Type	Description
details	string	Information about the failure.

skillPromptPlayFailed

This event indicates that a skill prompt play failed.

Name	Type	Description
details	string	Information about the failure.

skillPromptStopFailed

This event indicates that a skill prompt stop failed.

Name	Type	Description
details	string	Information about the failure.

skillPromptsChanged

This event, which contains no parameters, indicates that skill prompts were changed.

skillPromptsPermissionChanged

This event, which contains no parameters, indicates that skill prompts permission were changed.

skillsChanged

This event indicates that skills were changed. Use to select active skills.

Name	Type	Description
skills	skillInfo [0..unbounded]	Information about the skills.

skillsInitialized

This event indicates that skills were initialized.

Name	Type	Description
activeSkills	long [0..unbounded]	Initialized skills.
skills	skillInfo [0..unbounded]	Information about the skills.

skillsStatusChanged

This event indicates that the status of skills was changed. No action is required.

Name	Type	Description
skills	skillInfo [0..unbounded]	Information about the skills.

softphoneError

This event indicates that an error occurred during softphone initialization.

Name	Type	Description
message	string	Information about the error.

softphoneVolumeChanged

This event indicates that the softphone volume was changed.

Name	Type	Description
microphone	boolean	Whether the microphone volume was changed.
speaker	boolean	Whether the speaker volume was changed.
value	int	Value of the volume.

stationDisconnected

This event, which contains no parameters, indicates that the station has been disconnected.

switchToBDC

This event indicates a switch to the backup data center (BDC).

Name	Type	Description
reason	string	Reason for the switch.

switchToPDC

This event indicates a switch to the primary data center (PDC).

Name	Type	Description
reason	string	Reason for the switch.

timeAdjustmentChanged

This event indicates that the time adjustment was changed.

Name	Type	Description
timeAdjustment	long	Time adjustment value.

timeout

This event indicates that the time-out was reached during checkEvent.

Name	Type	Description
operation	string	

transferCallDisconnected

This event indicates that a transferred call was disconnected.

Name	Type	Description
callSessionId	string	Unique call identifier.

transferCompleted

This event indicates that a call transfer was completed.

Name	Type	Description
callSessionId	string	Unique call identifier.

transferError

This event indicates that an error occurred during a call transfer.

Name	Type	Description
callSessionId	string	Unique call identifier.
cause	string	Cause of the error.
errorCode	int	Numeric error code.

transferStarted

This event indicates that a call transfer was started.

Name	Type	Description
callSessionId	string	Unique call identifier.

updateStatus

This event indicates that a status was updated.

Name	Type	Description
status	agentBridgeStatus	Information about an agent's VCC session.

userUpdate

This event indicates that user data was updated.

Name	Type	Description
flags	long	

voicemailAccepted

This event indicates that a skill voicemail message was accepted.

Name	Type	Description
callSessionId	string	Unique call identifier.
callSkillId	long	Skill ID.
callType	int	Numeric value for the type of call.
campaignId	long	Campaign ID.
created	long	Date and time when the voicemail message was created.
customerTitle	string	Customer's title.
holdTime	long	
initialComments	string	Initial comments saved.
length	long	Length of comments.
offerId	long	

Name	Type	Description
ourNumber	string	Agent's phone number.
queueTime	long	Duration of voicemail message in the queue.
remoteNumber	string	Caller's phone number.
transferModule	string	User who transferred the voicemail message.
transferUserId	long	ID of the user who transferred the message.
voicemailId	long	ID of the voicemail message.

voicemailAdded

This event indicates that a personal voicemail message was added.

Name	Type	Description
value	recordingValue	Information about a recording.

voicemailCRMRecordChanged

This event indicates that a CRM record exists for the current voicemail message.

Name	Type	Description
record	map	Information about a mapping.
voicemailId	long	ID of the voicemail message.

voicemailDelivered

This event indicates that a skill voicemail message was delivered.

Name	Type	Description
campaignId	long	Campaign ID.
created	long	Date and time when the voicemail message was created.
initialComments	string	Initial comments saved.
length	long	Length of comments.
remoteNumber	string	Caller's phone number.
voicemailId	long	ID of the voicemail message.

voicemailEnded

This event indicates that a skill voicemail message ended.

Name	Type	Description
voicemailId	long	ID of the voicemail message.

voicemailPostponed

This event indicates that a voicemail message was postponed.

Name	Type	Description
voicemailId	long	ID of the voicemail message.

voicemailResumed

This event indicates that a voicemail message was resumed.

Name	Type	Description
voicemailId	long	ID of the voicemail message.

voicemailStateChanged

This event indicates that a voicemail message state was changed.

Name	Type	Description
callSessionId	string	Unique call identifier.
callSkillId	long	Skill ID.
callType	int	Numeric value for the type of call.
campaignId	long	Campaign ID.
created	long	Date and time when the voicemail message was created.
customerTitle	string	Customer's title.
holdTime	long	
initialComments	string	Initial comments saved.
length	long	Length of comments.
offerId	long	
ourNumber	string	Agent's phone number.
queueTime	long	Duration of voicemail message in the queue.
remoteNumber	string	Caller's phone number.
transferModule	string	User who transferred the voicemail message.
transferUserId	long	ID of the user who transferred the message.

Name	Type	Description
voicemailId	long	ID of the voicemail message.

Chapter 6

Exceptions

All error messages contain at least the message parameter, which is a string that describes the exception. Some exceptions contain additional parameters, which are described in the table.

Name	Description						
CrmRecordNotExistFault	The specified CRM contact record does not exist.						
DuplicateCrmRecordFault	More than one CRM contact record with the same name exists.						
	<table><tr><th>Name</th><th>Type</th><th>Description</th></tr><tr><td>crmRecord</td><td>crmRecord</td><td>Name of the duplicate record.</td></tr></table>	Name	Type	Description	crmRecord	crmRecord	Name of the duplicate record.
Name	Type	Description					
crmRecord	crmRecord	Name of the duplicate record.					
InvalidCrmFieldFault	The specified field is invalid.						
	<table><tr><th>Name</th><th>Type</th><th>Description</th></tr><tr><td>field</td><td>string</td><td>Name of invalid CRM record.</td></tr></table>	Name	Type	Description	field	string	Name of invalid CRM record.
Name	Type	Description					
field	string	Name of invalid CRM record.					
InvalidCrmRecordFault	The specified CRM record is invalid.						
ServiceFault	Web Services server error.						
	<table><tr><th>Name</th><th>Type</th><th>Description</th></tr><tr><td>code</td><td>string</td><td>Code that describes the error.</td></tr></table>	Name	Type	Description	code	string	Code that describes the error.
Name	Type	Description					
code	string	Code that describes the error.					